International Journal of Heat and Mass Transfer 105 (2017) 741–757

Contents lists available at ScienceDirect

International Journal of Heat and Mass Transfer

journal homepage: www.elsevier.com/locate/ijhmt

A simplified thermal lattice Boltzmann method without evolution of distribution functions

Z. Chen, C. Shu *, D. Tan

Department of Mechanical Engineering, National University of Singapore, 10 Kent Ridge Crescent, Singapore 119260, Singapore

ARTICLE INFO

Article history:
Received 31 March 2016
Received in revised form 14 July 2016
Accepted 8 October 2016

Keywords:
Chapman–Enskog expansion analysis
Simplified thermal lattice Boltzmann method
Lattice Boltzmann equation
Stability analysis
Thermal flows

ABSTRACT

In this paper, a simplified thermal lattice Boltzmann method (STLBM) without evolution of the distribution functions is developed for simulating incompressible thermal flows. With the assistance of the fractional step technique, the macroscopic governing equations recovered from Chapman–Enskog (C–E) expansion analysis are resolved through a predictor–corrector scheme. Then in both the predictor and corrector steps, using the isentropic properties of lattice tensors and relationships of C–E analysis, the macroscopic flow variables are explicitly calculated from the equilibrium and non-equilibrium distribution functions. In STLBM, the equilibrium distribution functions are calculated from the macroscopic variables, while the non-equilibrium distribution functions are evaluated from the differences between two equilibrium distribution functions at different locations and time levels. Therefore, STLBM directly updates the macroscopic variables during the computational process, which lowers the virtual memory cost and facilitates the implementation of physical boundary conditions. Through von Neumann stability analysis, the present method is proven to be unconditionally stable, which is further validated by numerical tests. Three representative examples are presented to demonstrate the robustness of STLBM in practical simulations and its flexibility on different types of meshes and boundaries.

© 2016 Elsevier Ltd. All rights reserved.

1. Introduction

In recent decades, the lattice Boltzmann method (LBM) has been continuously developed [1–4] and applied to various kinds of problems [5–11]. As a mesoscopic method, LBM offers an alternate approach to study the fluid problems, and is welcome among researchers due to its simplicity and explicitness. Based on kinetic theory, LBM solves the lattice Boltzmann equation (LBE) by tracking the evolution of the distribution function, which consists of two simple processes: "streaming" and "collision". Through Chapman–Enskog (C–E) expansion analysis, LBE can be proven to recover the macroscopic Navier–Stokes (N-S) equations. The relevant macroscopic properties can be easily obtained from the moments of the distribution functions.

Effective simulation of thermal flows has always been a popular topic in Computational Fluid Dynamics (CFD) due to its value both in academic explorations and in engineering applications [12–18]. As an advanced CFD method, the lattice Boltzmann method has also been extended to this topic [19–23]. Unlike isothermal flows, the effects of the temperature/energy are considered in simulating thermal flows. Therefore, various LBM models were proposed to incorporate the evolution of the temperature/energy into the computational process, which can be categorized into three approaches: the multispeed model [24,25], the passive scalar approach [26,27] and the thermal energy distribution model [28–32]. Among these models, the thermal energy distribution model is most recently developed, and has attracted much attention due to its better numerical stability. In this model, the thermal effects are incorporated into the computation by the evolution of the internal energy distribution function. Recently, Peng et al. [33] further simplified the thermal energy distribution model by neglecting the compression effect and heat dissipation. Such simplifications are robust in thermal simulations at the incompressible limit. The thermal energy distribution function model [30,34,35] remains the only method able to simulate incompressible viscous flows with imposed heat flux at the boundary (Neumann boundary condition).

The thermal LBM inherits the merits of the standard isothermal LBM but, at the same time, also suffers from its drawbacks. Firstly, due to the lattice uniformity, the thermal LBM is also only applicable on uniform meshes. Extra computational efforts are required to

* Corresponding author.
E-mail address: mpeshuc@nus.edu.sg (C. Shu).

http://dx.doi.org/10.1016/j.ijheatmasstransfer.2016.10.032
0017-9310/© 2016 Elsevier Ltd. All rights reserved.

apply it on non-uniform meshes. Secondly, the thermal LBM [28,33] requires large amounts of virtual memory, because the density and the internal energy distribution functions along all lattice velocity directions and at every grid point need to be stored during the computation. Another drawback is that the physical boundary conditions need to be transformed into the conditions of the distribution functions, which is a common problem for methods developed within LBM framework. Such transformations are quite challenging for cases with curved boundaries. In addition, constrained by numerical instability, simulating cases with high Reynolds number or high Rayleigh number is quite challenging for standard LBM [36]. To overcome these shortcomings, Wang et al. [37,38] combined the lattice Boltzmann solver and conventional Navier–Stokes solver, and proposed a novel thermal lattice Boltzmann flux solver (TLBFS). In TLBFS, FVM is applied globally to solve the macroscopic governing equations, while LBE solutions are reconstructed through C–E analysis at each cell interface to recover macroscopic fluxes. One important contribution of TLBFS is its treatment of the non-equilibrium distribution function, which is approximated by the difference between two equilibrium distribution functions at different locations and time levels. This solver has been proven to be both accurate and efficient, with easy implementation of boundary conditions. However, it should be noted that TLBFS involves two sets of solvers, which makes it inconvenient for practical applications. This flaw motivates us to develop the present simplified thermal lattice Boltzmann method which retains the merits of TLBFS but only requires a single solver.

In this paper, we propose a simplified thermal lattice Boltzmann method (STLBM) for simulating incompressible thermal flows. Based on the Chapman–Enskog expansion analysis, the LBE solutions can satisfy the macroscopic governing equations with the second-order of accuracy in space. By applying fractional step method, the recovered macroscopic governing equations can be solved through a predictor–corrector scheme. In STLBM, the formulations in the predictor–corrector steps are reconstructed from lattice properties and the relationships given by C–E analysis, which only involve equilibrium and non-equilibrium distribution functions. Specifically, the equilibrium distribution functions are obtained from the macroscopic flow variables; and the non-equilibrium distribution function is simply approximated by the difference between two equilibrium distribution functions at different locations and time levels. Therefore, unlike the standard thermal lattice Boltzmann method which tracks the evolution of the distribution functions, the present STLBM directly updates the macroscopic variables. Accordingly, without storing the distribution functions, STLBM presents a remarkable saving in virtual memory. Additionally, the physical boundary conditions can be implemented directly. Finally, the simplified lattice Boltzmann method shows nice performance in terms of numerical stability, and can be analytically proven to be unconditionally stable.

The rest of the paper is organized as follows. In Section 2, the basic macroscopic governing equations, the thermal lattice Boltzmann method, and the Chapman–Enskog expansion analysis that links these two systems are briefly introduced. Section 3 gives detailed derivation process, the von Neumann stability analysis and the boundary treatment of the simplified thermal lattice Boltzmann method proposed in this paper. Section 4 presents three representative numerical examples (natural convection in a square cavity, the natural convection in a concentric annulus and the mixed heat transfer from a heated circular cylinder) to validate the convergence, stability, and accuracy of the present method, as well as its adaptability on non-uniform meshes with curved boundaries. The numerical results are compared with reference data in literature. Brief conclusions are finally drawn in Section 5.

2. Macroscopic governing equations and thermal lattice Boltzmann method

2.1. Macroscopic governing equations

For general thermal viscous flows, the macroscopic governing equations include the following continuity, momentum and energy equations [39]

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \rho \mathbf{u} = 0 \tag{1}$$

$$\frac{\partial \rho \mathbf{u}}{\partial t} + \nabla \cdot (\rho \mathbf{u}\mathbf{u}) = -\nabla p + \upsilon \nabla \cdot \left[\nabla \rho \mathbf{u} + (\nabla \rho \mathbf{u})^T\right] + \mathbf{F}_E \tag{2}$$

$$\frac{\partial}{\partial t}(\rho e) + \nabla \cdot (\rho \mathbf{u} e) = \chi \nabla^2 (\rho e) \tag{3}$$

where ρ, $\mathbf{u}$, p and υ denote the density, the velocity vector, the pressure and the kinematic viscosity, respectively. $\mathbf{F}_E$ is the external force term; χ represents the thermal diffusivity; and e is the internal energy defined as $e = DRT/2$. D, R and T are the number of spatial dimensions, the gas constant and the temperature, respectively. In the present work, we set $D = 2$ and $R = 1$ to maintain consistency with previous studies [37].

The heat transfer can be conducted through conduction, convection and radiation [40]. Here, we mainly focus on the heat convection process in the fluid. Forced convection and natural convection are the two major heat convection problems. For forced convection, the bulk motion of the fluid is driven by external means, e.g., fans or pumps. Therefore, it is usually assumed that the internal energy, or the temperature, has no effect on the velocity field (momentum equation). For natural convection, the motion of the fluid is driven by the buoyancy force, which is related to the temperature difference. Consequently, the momentum and energy equations have to be coupled, and the velocity and the temperature affect each other.

2.2. Thermal lattice Boltzmann method

The thermal lattice Boltzmann model with BGK approximation [28,33] can be expressed as

$$f_\alpha(\mathbf{r} + \mathbf{e}_\alpha \delta_t, t + \delta_t) = f_\alpha(\mathbf{r}, t) + \frac{f_\alpha^{eq}(\mathbf{r}, t) - f_\alpha(\mathbf{r}, t)}{\tau_\upsilon}, \quad \alpha = 0, 1, \ldots, M \tag{4}$$

$$g_\alpha(\mathbf{r} + \mathbf{e}_\alpha \delta_t, t + \delta_t) = g_\alpha(\mathbf{r}, t) + \frac{g_\alpha^{eq}(\mathbf{r}, t) - g_\alpha(\mathbf{r}, t)}{\tau_c}, \quad \alpha = 0, 1, \ldots, N \tag{5}$$

where f_α and g_α are the density distribution function and the internal energy distribution function along the α direction, respectively; The superscript "eq" denotes the equilibrium state; τ_υ and τ_c are single relaxation parameters related to the kinematic viscosity and the thermal diffusivity, respectively; δ_t is the time step applied in the model; M and N are the number of the lattice velocities used in the density and the internal energy distribution functions respectively. In thermal LBM, the macroscopic properties can be obtained from the conservation laws of mass, momentum and energy:

$$\rho = \sum_\alpha f_\alpha, \quad \rho \mathbf{u} = \sum_\alpha f_\alpha \mathbf{e}_\alpha, \quad \rho e = \sum_\alpha g_\alpha \tag{6}$$

The thermal lattice Boltzmann method is associated with the lattice velocity model. In the present work, the 9-bit lattice velocity model (D2Q9) is applied (see Fig. 1) [28,41,42], which consists of the following lattice velocity directions

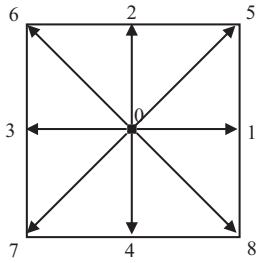


Fig. 1. Illustration of the D2Q9 model.

$$\mathbf{e}_\alpha = \begin{cases} 0 & \alpha = 0 \\ (\pm 1, 0), (0, \pm 1) & \alpha = 1, 2, 3, 4 \\ (\pm 1, \pm 1) & \alpha = 5, 6, 7, 8 \end{cases} \tag{7}$$

The equilibrium distribution functions are given by

$$f_\alpha^{eq} = \rho \omega_\alpha \left[1 + \frac{\mathbf{e}_\alpha \cdot \mathbf{u}}{c_s^2} + \frac{(\mathbf{e}_\alpha \cdot \mathbf{u})^2 - (c_s |\mathbf{u}|)^2}{2 c_s^4} \right], \quad \alpha = 0, 1, \ldots, 8 \tag{8}$$

$$g_\alpha^{eq}(\mathbf{r}, t) = \begin{cases} -\frac{2\rho e}{3} \frac{|\mathbf{u}|^2}{c^2} & \alpha = 0 \\ \frac{\rho e}{9} \left[\frac{3}{2} + \frac{3}{2} \frac{\mathbf{e}_\alpha \cdot \mathbf{u}}{c^2} + \frac{9}{2} \frac{(\mathbf{e}_\alpha \cdot \mathbf{u})^2}{c^4} - \frac{3}{2} \frac{|\mathbf{u}|^2}{c^2} \right] & \alpha = 1 - 4 \\ \frac{\rho e}{36} \left[3 + 6 \cdot \frac{\mathbf{e}_\alpha \cdot \mathbf{u}}{c^2} + \frac{9}{2} \frac{(\mathbf{e}_\alpha \cdot \mathbf{u})^2}{c^4} - \frac{3}{2} \frac{|\mathbf{u}|^2}{c^2} \right] & \alpha = 5 - 8 \end{cases} \tag{9}$$

where $c = \delta_x / \delta_t$ is set to be 1 in this work; and δ_x is the lattice spacing. The weighting coefficient ω_α and the sound speed c_s are

$$\omega_0 = \frac{4}{9} \quad \omega_{1-4} = \frac{1}{9} \quad \omega_{5-8} = \frac{1}{36} \quad c_s = \frac{c}{\sqrt{3}}. \tag{10}$$

2.3. Chapman–Enskog expansion analysis

The Chapman–Enskog (C–E) expansion analysis offers the link connecting the macroscopic governing Eqs. (1)–(3) and the mesoscopic lattice Boltzmann Eqs. (4) and (5) [43,44]. Basically, the foundation of C–E analysis is the multi-scale expansion, with which the density distribution function, the internal energy distribution function, the temporal derivative and the spatial derivative can be written as

$$f_\alpha = f_\alpha^{(0)} + \varepsilon f_\alpha^{(1)} + \varepsilon^2 f_\alpha^{(2)} \tag{11}$$

$$g_\alpha = g_\alpha^{(0)} + \varepsilon g_\alpha^{(1)} + \varepsilon^2 g_\alpha^{(2)} \tag{12}$$

$$\frac{\partial}{\partial t} = \varepsilon \frac{\partial}{\partial t_0} + \varepsilon^2 \frac{\partial}{\partial t_1} \tag{13}$$

$$\nabla_r = \varepsilon \nabla_{r1} \tag{14}$$

where ε is a small perturbation number proportional to the Knudsen number [45].

By using Taylor series expansion with the second-order of accuracy in space and time, Eqs. (4) and (5) can be approximated by

$$\left(\frac{\partial}{\partial t} + \mathbf{e}_\alpha \cdot \nabla \right) f_\alpha + \frac{\delta_t}{2} \left(\frac{\partial}{\partial t} + \mathbf{e}_\alpha \cdot \nabla \right)^2 f_\alpha + \frac{1}{\tau \delta_t} (f_\alpha - f_\alpha^{eq}) + O\left(\delta_t^2\right) = 0 \tag{15}$$

$$\left(\frac{\partial}{\partial t} + \mathbf{e}_\alpha \cdot \nabla \right) g_\alpha + \frac{\delta_t}{2} \left(\frac{\partial}{\partial t} + \mathbf{e}_\alpha \cdot \nabla \right)^2 g_\alpha + \frac{1}{\tau \delta_t} (g_\alpha - g_\alpha^{eq}) + O\left(\delta_t^2\right) = 0 \tag{16}$$

Substituting Eqs. (11)–(14) into Eqs. (15) and (16), and applying the conservation laws of mass, momentum and energy, gives the following equations

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \left(\sum_\alpha \mathbf{e}_\alpha f_\alpha^{eq} \right) = 0 \tag{17}$$

$$\frac{\partial \rho \mathbf{u}}{\partial t} + \nabla \cdot \Pi = 0 \tag{18}$$

$$\frac{\partial \rho e}{\partial t} + \nabla \cdot \mathbf{Q} = 0 \tag{19}$$

where Π and $\mathbf{Q}$ are the momentum flux tensor and the thermal flux vector in the following form

$$\Pi_{\beta\gamma} = \sum_\alpha (\mathbf{e}_\alpha)_\beta (\mathbf{e}_\alpha)_\gamma \left[f_\alpha^{eq} + \left(1 - \frac{1}{2\tau_\upsilon} \right) \varepsilon f_\alpha^{(1)} \right] \tag{20}$$

$$\mathbf{Q} = \sum_\alpha \mathbf{e}_\alpha \left[g_\alpha^{eq} + \left(1 - \frac{1}{2\tau_c} \right) \varepsilon g_\alpha^{(1)} \right] \tag{21}$$

Here, $(\mathbf{e}_\alpha)_\beta$ and $(\mathbf{e}_\alpha)_\gamma$ are the components of the lattice velocity vector $\mathbf{e}_\alpha$ in β-direction and in γ-direction, respectively. It can be shown that the distribution functions f_α and g_α are approximated to the first order of ε by $f_\alpha^{eq} + \varepsilon f_\alpha^{(1)}$ and $g_\alpha^{eq} + \varepsilon g_\alpha^{(1)}$, respectively. $\varepsilon f_\alpha^{(1)}$ and $\varepsilon g_\alpha^{(1)}$ can therefore be written as

$$f_\alpha^{neq} = f_\alpha - f_\alpha^{eq} \approx \varepsilon f_\alpha^{(1)} = -\tau_\upsilon \delta_t \left(\frac{\partial}{\partial t} + \mathbf{e}_\alpha \cdot \nabla \right) f_\alpha^{eq} = -\tau_\upsilon \delta_t \mathrm{D} f_\alpha^{eq} \tag{22}$$

$$g_\alpha^{neq} = g_\alpha - g_\alpha^{eq} \approx \varepsilon g_\alpha^{(1)} = -\tau_c \delta_t \left(\frac{\partial}{\partial t} + \mathbf{e}_\alpha \cdot \nabla \right) g_\alpha^{eq} = -\tau_c \delta_t \mathrm{D} g_\alpha^{eq} \tag{23}$$

where the superscript "*neq*" represents the non-equilibrium distribution function; and the operator D denotes $D = \left(\frac{\partial}{\partial t} + \mathbf{e}_\alpha \cdot \nabla \right)$. By using Eqs. (22) and (23), Eqs. (20) and (21) can be written as

$$\Pi_{\beta\gamma} = \sum_\alpha (\mathbf{e}_\alpha)_\beta (\mathbf{e}_\alpha)_\gamma \left[f_\alpha^{eq} + \left(1 - \frac{1}{2\tau_\upsilon} \right) f_\alpha^{neq} \right] \tag{24}$$

$$\mathbf{Q} = \sum_\alpha \mathbf{e}_\alpha \left[g_\alpha^{eq} + \left(1 - \frac{1}{2\tau_c} \right) f_\alpha^{neq} \right] \tag{25}$$

Note that to fulfill the conservation laws, the non-equilibrium distribution functions have the following relationships:

$$\sum_\alpha f_\alpha^{neq} = 0 \quad \sum_\alpha \mathbf{e}_\alpha f_\alpha^{neq} = 0 \quad \sum_\alpha g_\alpha^{neq} = 0 \tag{26}$$

With the conservation laws and the above relationships of the non-equilibrium distribution functions, Eqs. (17)–(19) can recover the macroscopic governing Eqs. (1)–(3) for thermal flows with the following relationships between the relaxation parameters (τ_υ & τ_c) and the kinematic viscosity υ & thermal diffusivity χ :

$$\upsilon = c_s^2 \delta_t \left(\tau_\upsilon - \frac{1}{2} \right) \tag{27}$$

$$\chi = \frac{2}{3} c^2 \delta_t \left(\tau_c - \frac{1}{2} \right) \tag{28}$$

Here we are neglecting the effect of inconsistency of kinematic viscosity into energy balance equation discussed by He et al. [28].

3. A simplified thermal lattice Boltzmann method (STLBM) without evolution of distribution functions

3.1. Formulations of STLBM

In this section, the simplified thermal lattice Boltzmann method (STLBM) without evolution of the distribution functions is

presented. It is shown in the previous section that C–E analysis recovers the following macroscopic governing equations

$$\frac{\partial \mathbf{W}}{\partial t}+\frac{\partial \mathbf{E}}{\partial x}+\frac{\partial \mathbf{F}}{\partial y}+\mathbf{S}=0 \tag{29}$$

where

$$\mathbf{W}=\begin{Bmatrix} W_1\\ W_2\\ W_3\\ W_4\end{Bmatrix}=\begin{Bmatrix}\rho\\ \rho u\\ \rho v\\ \rho e\end{Bmatrix},\quad \mathbf{E}=\begin{Bmatrix}P_x\\ \Pi_{xx}\\ \Pi_{yx}\\ Q_x\end{Bmatrix},\quad \mathbf{F}=\begin{Bmatrix}P_y\\ \Pi_{xy}\\ \Pi_{yy}\\ Q_y\end{Bmatrix},\quad \mathbf{S}=\begin{Bmatrix}0\\ -F_{Ex}\\ -F_{Ey}\\ 0\end{Bmatrix} \tag{30}$$

$$P_x=\sum_\alpha e_{\alpha x}f_\alpha^{eq}\quad P_y=\sum_\alpha e_{\alpha y}f_\alpha^{eq} \tag{31}$$

$$\begin{aligned}\Pi_{xx}&=\sum_\alpha e_{\alpha x}e_{\alpha x}\left[f_\alpha^{eq}+\left(1-\tfrac{1}{2\tau_v}\right)f_\alpha^{neq}\right]\quad \Pi_{xy}=\sum_\alpha e_{\alpha x}e_{\alpha y}\left[f_\alpha^{eq}+\left(1-\tfrac{1}{2\tau_v}\right)f_\alpha^{neq}\right]\\ \Pi_{yx}&=\sum_\alpha e_{\alpha y}e_{\alpha x}\left[f_\alpha^{eq}+\left(1-\tfrac{1}{2\tau_v}\right)f_\alpha^{neq}\right]\quad \Pi_{yy}=\sum_\alpha e_{\alpha y}e_{\alpha y}\left[f_\alpha^{eq}+\left(1-\tfrac{1}{2\tau_v}\right)f_\alpha^{neq}\right]\end{aligned} \tag{32}$$

$$\begin{aligned}Q_x&=\sum_\alpha e_{\alpha x}\left[g_\alpha^{eq}+\left(1-\frac{1}{2\tau_c}\right)g_\alpha^{neq}\right]\\ Q_y&=\sum_\alpha e_{\alpha y}\left[g_\alpha^{eq}+\left(1-\frac{1}{2\tau_c}\right)g_\alpha^{neq}\right]\end{aligned} \tag{33}$$

Assisted by the fractional step technique [46], the above Eq. (29) can be solved in the following two steps:

Predictor step:

$$\frac{\partial \rho}{\partial t}+\nabla\cdot\left(\sum_\alpha \mathbf{e}_\alpha f_\alpha^{eq}\right)=0 \tag{34}$$

$$\frac{\partial \rho \mathbf{u}}{\partial t}+\nabla\cdot\left[\sum_\alpha (\mathbf{e}_\alpha)_\beta(\mathbf{e}_\alpha)_\gamma f_\alpha^{eq}+\frac{1}{2\tau_v}\sum_\alpha (\mathbf{e}_\alpha)_\beta(\mathbf{e}_\alpha)_\gamma f_\alpha^{neq}\right]=0 \tag{35}$$

$$\frac{\partial \rho e}{\partial t}+\nabla\cdot\left[\sum_\alpha \mathbf{e}_\alpha g_\alpha^{eq}+\frac{1}{2\tau_c}\sum_\alpha \mathbf{e}_\alpha g_\alpha^{neq}\right]=0 \tag{36}$$

Corrector step:

$$\frac{\partial \rho}{\partial t}=0 \tag{37}$$

$$\frac{\partial \rho \mathbf{u}}{\partial t}+\nabla\cdot\left[\left(1-\frac{1}{\tau_v}\right)\sum_\alpha (\mathbf{e}_\alpha)_\beta(\mathbf{e}_\alpha)_\gamma f_\alpha^{neq}\right]-\mathbf{F}_E=0 \tag{38}$$

$$\frac{\partial \rho e}{\partial t}+\nabla\cdot\left[\left(1-\frac{1}{\tau_c}\right)\sum_\alpha \mathbf{e}_\alpha g_\alpha^{neq}\right]=0 \tag{39}$$

The solution of the equations above can be reconstructed in the lattice Boltzmann frame with the use of lattice properties. Specifically, in the predictor step, the intermediate density, velocity and internal energy can be estimated as

$$\rho^*=\sum_\alpha f_\alpha^{eq}(\mathbf{r}-\mathbf{e}_\alpha\delta_t,t-\delta_t) \tag{40}$$

$$\rho^*\mathbf{u}^*=\sum_\alpha \mathbf{e}_\alpha f_\alpha^{eq}(\mathbf{r}-\mathbf{e}_\alpha\delta_t,t-\delta_t) \tag{41}$$

$$\rho^* e^*=\sum_\alpha g_\alpha^{eq}(\mathbf{r}-\mathbf{e}_\alpha\delta_t,t-\delta_t) \tag{42}$$

where the superscript $*$ indicates the intermediate value. It can be proven that the above three equations are equivalent to Eqs. (34)–(36) with the second-order accuracy in both time and space. Using a Taylor-series expansion with the third-order truncated error in space and time, the equilibrium density and internal energy distribution functions can be approximated by

$$\begin{aligned}f_\alpha^{eq}(\mathbf{r}-\mathbf{e}_\alpha\delta_t,t-\delta_t)&=f_\alpha^{eq}(\mathbf{r},t)-\delta_t \mathrm{D}f_\alpha^{eq}(\mathbf{r},t)+\tfrac{\delta_t^2}{2}\mathrm{D}^2 f_\alpha^{eq}(\mathbf{r},t)+O(\delta_t^3)\\ &=f_\alpha^{eq}(\mathbf{r},t)-\delta_t \mathrm{D}f_\alpha^{eq}(\mathbf{r},t)-\tfrac{\delta_t}{2\tau_v}\mathrm{D}f_\alpha^{neq}(\mathbf{r},t)+O(\delta_t^3)\end{aligned} \tag{43}$$

$$\begin{aligned}g_\alpha^{eq}(\mathbf{r}-\mathbf{e}_\alpha\delta_t,t-\delta_t)&=g_\alpha^{eq}(\mathbf{r},t)-\delta_t \mathrm{D}g_\alpha^{eq}(\mathbf{r},t)+\tfrac{\delta_t^2}{2}\mathrm{D}^2 g_\alpha^{eq}(\mathbf{r},t)+O(\delta_t^3)\\ &=g_\alpha^{eq}(\mathbf{r},t)-\delta_t \mathrm{D}g_\alpha^{eq}(\mathbf{r},t)-\tfrac{\delta_t}{2\tau_c}\mathrm{D}g_\alpha^{neq}(\mathbf{r},t)+O(\delta_t^3)\end{aligned} \tag{44}$$

Note that Eqs. (22) and (23) are involved in deriving the above equations. Substituting Eq. (43) into Eq. (40) yields:

$$\begin{aligned}\rho^*=&\sum_\alpha f_\alpha^{eq}(\mathbf{r},t)-\delta_t\left[\frac{\partial}{\partial t}\sum_\alpha f_\alpha^{eq}(\mathbf{r},t)+\nabla\cdot\sum_\alpha \mathbf{e}_\alpha f_\alpha^{eq}(\mathbf{r},t)\right]\\ &-\frac{\delta_t}{2\tau_v}\left[\frac{\partial}{\partial t}\sum_\alpha f_\alpha^{neq}(\mathbf{r},t)+\nabla\cdot\sum_\alpha \mathbf{e}_\alpha f_\alpha^{neq}(\mathbf{r},t)\right]+O(\delta_t^3)\end{aligned} \tag{45}$$

Note that from the conservation laws in Eq. (6), the first term on the Right-Hand-Side (RHS) of Eq. (45) is the density at $(\mathbf{r}, t)$, which could be considered as ρ^*. According to the features of the non-equilibrium distribution functions in Eq. (26), the third term on RHS vanishes. Overall, Eq. (45) can be simplified into

$$\frac{\partial \rho}{\partial t}+\nabla\cdot\left(\sum_\alpha \mathbf{e}_\alpha f_\alpha^{eq}\right)+O(\delta_t^2)=0 \tag{46}$$

The above equation shows that Eq. (40) can recover Eq. (34) with second-order accuracy in space and time.

Similarly, substituting Eqs. (43) and (44) into Eqs. (41) and (42) respectively gives

$$\begin{aligned}\rho^*\mathbf{u}^*=&\sum_\alpha \mathbf{e}_\alpha f_\alpha^{eq}(\mathbf{r},t)\\ &-\delta_t\left[\frac{\partial}{\partial t}\sum_\alpha \mathbf{e}_\alpha f_\alpha^{eq}(\mathbf{r},t)+\nabla\cdot\sum_\alpha (\mathbf{e}_\alpha)_\beta(\mathbf{e}_\alpha)_\gamma f_\alpha^{eq}(\mathbf{r},t)\right]\\ &-\frac{\delta_t}{2\tau_v}\sum_\alpha \mathbf{e}_\alpha \mathrm{D}f_\alpha^{neq}(\mathbf{r},t)+O(\delta_t^3)\end{aligned} \tag{47}$$

$$\begin{aligned}\rho^* e^*=&\sum_\alpha g_\alpha^{eq}(\mathbf{r},t)-\delta_t\left[\frac{\partial}{\partial t}\sum_\alpha g_\alpha^{eq}(\mathbf{r},t)+\nabla\cdot\sum_\alpha \mathbf{e}_\alpha g_\alpha^{eq}(\mathbf{r},t)\right]\\ &-\frac{\delta_t}{2\tau_c}\sum_\alpha \mathrm{D}g_\alpha^{neq}(\mathbf{r},t)+O(\delta_t^3)\end{aligned} \tag{48}$$

According to the conservation laws (Eq. (6)), the first terms on RHS of Eqs. (47) and (48) can be considered as $\rho^*\mathbf{u}^*$ and $\rho^* e^*$, respectively. The third terms on RHS of Eqs. (47) and (48) can be simplified by the features of the non-equilibrium distribution functions (Eq. (26)) to

$$\sum_\alpha \mathbf{e}_\alpha \mathrm{D}f_\alpha^{neq}(\mathbf{r},t)=\nabla\cdot\left[\sum_\alpha (\mathbf{e}_\alpha)_\beta(\mathbf{e}_\alpha)_\gamma f_\alpha^{neq}(\mathbf{r},t)\right] \tag{49}$$

$$\sum_\alpha \mathrm{D}g_\alpha^{neq}(\mathbf{r},t)=\nabla\cdot\left[\sum_\alpha \mathbf{e}_\alpha g_\alpha^{neq}(\mathbf{r},t)\right] \tag{50}$$

Therefore, Eqs. (47) and (48) can be reduced to

$$\frac{\partial \rho \mathbf{u}}{\partial t}+\nabla\cdot\left[\sum_{\alpha}(\mathbf{e}_\alpha)_\beta(\mathbf{e}_\alpha)_\gamma f_\alpha^{eq}+\frac{1}{2\tau_\upsilon}\sum_{\alpha}(\mathbf{e}_\alpha)_\beta(\mathbf{e}_\alpha)_\gamma f_\alpha^{neq}\right]+O(\delta_t^2)=0 \tag{51}$$

$$\frac{\partial \rho e}{\partial t}+\nabla\cdot\left[\sum_{\alpha}\mathbf{e}_\alpha g_\alpha^{eq}+\frac{1}{2\tau_c}\sum_{\alpha}\mathbf{e}_\alpha g_\alpha^{neq}\right]+O(\delta_t^2)=0 \tag{52}$$

Eqs. (51) and (52) show that Eqs. (41) and (42) can recover Eqs. (35) and (36) with second-order accuracy in space and time.

In the corrector step, we wish to recover Eqs. (37)–(39) in the lattice Boltzmann frame. The corrected density, velocity and energy are evaluated as

$$\rho(\mathbf{r},t)=\rho^* \tag{53}$$

$$\rho(\mathbf{r},t)\mathbf{u}(\mathbf{r},t)=\rho^*\mathbf{u}^*+\mathbf{F}_E\delta_t+\left(1-\frac{1}{\tau_\upsilon}\right)\sum_{\alpha}\mathbf{e}_\alpha f_\alpha^{neq}(\mathbf{r}-\mathbf{e}_\alpha\delta_t,t) \tag{54}$$

$$\rho(\mathbf{r},t)e(\mathbf{r},t)=\rho^* e^*+\left(1-\frac{1}{\tau_c}\right)\sum_{\alpha}g_\alpha^{neq}(\mathbf{r}-\mathbf{e}_\alpha\delta_t,t) \tag{55}$$

Apparently, Eq. (53) recovers Eq. (37) in the corrector step exactly. Now we focus on the correction terms in Eqs. (54) and (55). By using Taylor series expansion, $f_\alpha^{neq}(\mathbf{r}-\mathbf{e}_\alpha\delta_t,t)$ and $g_\alpha^{neq}(\mathbf{r}-\mathbf{e}_\alpha\delta_t,t)$ can be approximated by

$$f_\alpha^{neq}(\mathbf{r}-\mathbf{e}_\alpha\delta_t,t)=f_\alpha^{neq}(\mathbf{r},t)-\delta_t\mathbf{e}_\alpha\cdot\nabla f_\alpha^{neq}(\mathbf{r},t)+\frac{\delta_t^2}{2}\left[\mathbf{e}_\alpha\cdot\nabla f_\alpha^{neq}(\mathbf{r},t)\right]^2+O(\delta_t^3) \tag{56}$$

$$g_\alpha^{neq}(\mathbf{r}-\mathbf{e}_\alpha\delta_t,t)=g_\alpha^{neq}(\mathbf{r},t)-\delta_t\mathbf{e}_\alpha\cdot\nabla g_\alpha^{neq}(\mathbf{r},t)+\frac{\delta_t^2}{2}\left[\mathbf{e}_\alpha\cdot\nabla g_\alpha^{neq}(\mathbf{r},t)\right]^2+O(\delta_t^3) \tag{57}$$

Consequently, the last terms in Eqs. (54) and (55) can be respectively manipulated as follows:

$$\left(1-\frac{1}{\tau_\upsilon}\right)\frac{1}{\delta_t}\sum_{\alpha}\mathbf{e}_\alpha f_\alpha^{neq}(\mathbf{r}-\mathbf{e}_\alpha\delta_t,t)=\left(1-\frac{1}{\tau_\upsilon}\right)\frac{1}{\delta_t}\sum_{\alpha}\mathbf{e}_\alpha f_\alpha^{neq}(\mathbf{r},t)-\left(1-\frac{1}{\tau_\upsilon}\right)\left[\nabla\cdot\sum_{\alpha}(\mathbf{e}_\alpha)_\beta(\mathbf{e}_\alpha)_\gamma f_\alpha^{neq}(\mathbf{r},t)-\frac{\delta_t}{2}\sum_{\alpha}\mathbf{e}_\alpha\left[\mathbf{e}_\beta\cdot\nabla f_\alpha^{neq}(\mathbf{r},t)\right]^2\right]+O(\delta_t^2) \tag{58}$$

$$\left(1-\frac{1}{\tau_c}\right)\frac{1}{\delta_t}\sum_{\alpha}g_\alpha^{neq}(\mathbf{r}-\mathbf{e}_\alpha\delta_t,t)=\left(1-\frac{1}{\tau_c}\right)\frac{1}{\delta_t}\sum_{\alpha}g_\alpha^{neq}(\mathbf{r},t)-\left(1-\frac{1}{\tau_c}\right)\left[\nabla\cdot\sum_{\alpha}\mathbf{e}_\alpha g_\alpha^{neq}(\mathbf{r},t)-\frac{\delta_t}{2}\sum_{\alpha}\left[\mathbf{e}_\alpha\cdot\nabla g_\alpha^{neq}(\mathbf{r},t)\right]^2\right]+O\left(\delta_t^2\right) \tag{59}$$

According to Eqs. (22) and (23), f_α^{neq} and g_α^{neq} are of the order δ_t. Using Eq. (26), Eqs. (58) and (59) can be further reduced to

$$\left(1-\frac{1}{\tau_\upsilon}\right)\frac{1}{\delta_t}\sum_{\alpha}\mathbf{e}_\alpha f_\alpha^{neq}(\mathbf{r}-\mathbf{e}_\alpha\delta_t,t)=-\nabla\cdot\left[\left(1-\frac{1}{\tau_\upsilon}\right)\sum_{\alpha}(\mathbf{e}_\alpha)_\beta(\mathbf{e}_\alpha)_\gamma f_\alpha^{neq}(\mathbf{r},t)\right]+O(\delta_t^2) \tag{60}$$

$$\left(1-\frac{1}{\tau_c}\right)\frac{1}{\delta_t}\sum_{\alpha}g_\alpha^{neq}(\mathbf{r}-\mathbf{e}_\alpha\delta_t,t)=-\nabla\cdot\left[\left(1-\frac{1}{\tau_c}\right)\sum_{\alpha}\mathbf{e}_\alpha g_\alpha^{neq}(\mathbf{r},t)\right]+O(\delta_t^2) \tag{61}$$

Obviously, Eqs. (60) and (61) recover the last terms in Eqs. (38) and (39) with second-order accuracy in space. Overall, Eqs. (54) and (55) can recover Eqs. (38) and (39) with second-order accuracy in space. According to the Boussinesq approximation [42], the external force term can be regarded as a buoyance force related to the temperature profile, i.e.,

$$\mathbf{F}_E=\begin{pmatrix}0\\-\rho g\beta(T-T_m)\end{pmatrix} \tag{62}$$

where g is the gravitational acceleration; β is the thermal expansion coefficient; and T_m is the average temperature. For ease in implementation, the basic formulations of STLBM are summarized as follows:

Predictor step:

$$\rho^*=\sum_{\alpha}f_\alpha^{eq}(\mathbf{r}-\mathbf{e}_\alpha\delta_t,t-\delta_t) \tag{63}$$

$$\rho^*\mathbf{u}^*=\sum_{\alpha}\mathbf{e}_\alpha f_\alpha^{eq}(\mathbf{r}-\mathbf{e}_\alpha\delta_t,t-\delta_t) \tag{64}$$

$$\rho^* e^*=\sum_{\alpha}g_\alpha^{eq}(\mathbf{r}-\mathbf{e}_\alpha\delta_t,t-\delta_t) \tag{65}$$

Corrector step:

$$\rho(\mathbf{r},t)=\rho^* \tag{66}$$

$$\rho(\mathbf{r},t)\mathbf{u}(\mathbf{r},t)=\rho^*\mathbf{u}^*+\left(1-\frac{1}{\tau_\upsilon}\right)\sum_{\alpha}\mathbf{e}_\alpha f_\alpha^{neq}(\mathbf{r}-\mathbf{e}_\alpha\delta_t,t)+\mathbf{F}_E\delta_t \tag{67}$$

$$\rho(\mathbf{r},t)e(\mathbf{r},t)=\rho^* e^*+\left(1-\frac{1}{\tau_c}\right)\sum_{\alpha}g_\alpha^{neq}(\mathbf{r}-\mathbf{e}_\alpha\delta_t,t) \tag{68}$$

One issue in the application of the above formulation is the calculation of the non-equilibrium distribution functions. Inspired by LBFS [47], and recalling Eqs. (22) and (23) given in C–E analysis, the non-equilibrium density distribution functions are approximated by

$$f_\alpha^{neq}(\mathbf{r},t)=-\tau_\upsilon\delta_t\mathrm{D}f_\alpha^{eq}(\mathbf{r},t)=-\tau_\upsilon\left[f_\alpha^{eq,*}(\mathbf{r},t)-f_\alpha^{eq}(\mathbf{r}-\mathbf{e}_\alpha\delta_t,t-\delta_t)\right] \tag{69}$$

$$g_\alpha^{neq}(\mathbf{r},t)=-\tau_c\delta_t\mathrm{D}g_\alpha^{eq}(\mathbf{r},t)=-\tau_c\left[g_\alpha^{eq,*}(\mathbf{r},t)-g_\alpha^{eq}(\mathbf{r}-\mathbf{e}_\alpha\delta_t,t-\delta_t)\right] \tag{70}$$

Note that the equilibrium distribution function $f_\alpha^{eq,*}(r,t)$ and $g_\alpha^{eq,*}(r,t)$ in the above equations are evaluated from the intermediate density, velocity and internal energy obtained in the predictor step.

3.2. Stability analysis

The von Neumann stability analysis [48] is a powerful tool to evaluate the stability performance of a new scheme. In this section, we conduct this analysis on the present simplified thermal lattice Boltzmann method. For the ease of analysis, the variables are denoted as

$$y_1=\rho,\quad y_2=\rho u,\quad y_3=\rho v,\quad y_4=\rho e \tag{71}$$

We also denote the vector of the variables as

$$\mathbf{Y}=(y_1\quad y_2\quad y_3\quad y_4)^{\mathrm{T}} \tag{72}$$

Substituting the above notation into the equilibrium density/internal energy distribution functions gives:

$$f_\alpha^{eq}=w_\alpha y_{1\alpha}+\frac{w_\alpha}{c_s^2}\left(e_{\alpha x}y_{2\alpha}+e_{\alpha y}y_{3\alpha}\right)+\frac{w_\alpha}{2c_s^4}\frac{\left(e_{\alpha x}y_{2\alpha}+e_{\alpha y}y_{3\alpha}\right)^2}{y_{1\alpha}}-\frac{w_\alpha}{2c_s^2}\frac{y_{2\alpha}^2+y_{3\alpha}^2}{y_{1\alpha}} \tag{73}$$

$$g_\alpha^{eq}=\begin{cases} -\frac{2}{3}y_{4\alpha}\frac{y_{2\alpha}^2+y_{3\alpha}^2}{y_{1\alpha}^2} & \alpha=0 \\ \frac{1}{6}y_{4\alpha}+\frac{1}{6}y_{4\alpha}\frac{e_{xx}y_{2\alpha}+e_{xy}y_{3\alpha}}{y_{1\alpha}}+\frac{1}{2}y_{4\alpha}\left(\frac{e_{xx}y_{2\alpha}+e_{xy}y_{3\alpha}}{y_{1\alpha}}\right)^2-\frac{1}{6}y_{4\alpha}\frac{y_{2\alpha}^2+y_{3\alpha}^2}{y_{1\alpha}^2} & \alpha=1-4 \\ \frac{1}{12}y_{4\alpha}+\frac{1}{6}y_{4\alpha}\frac{e_{xx}y_{2\alpha}+e_{xy}y_{3\alpha}}{y_{1\alpha}}+\frac{1}{8}y_{4\alpha}\left(\frac{e_{xx}y_{2\alpha}+e_{xy}y_{3\alpha}}{y_{1\alpha}}\right)^2-\frac{1}{24}y_{4\alpha}\frac{y_{2\alpha}^2+y_{3\alpha}^2}{y_{1\alpha}^2} & \alpha=5-8 \end{cases} \tag{74}$$

where $y_{1\alpha}, y_{2\alpha}, y_{3\alpha}$ and $y_{4\alpha}$ represent the functional values of y_1, y_2, y_3 and y_4 at the location ($\mathbf{r}-\mathbf{e}_\alpha$, δ_t), respectively. The above expressions can be substituted into the formulations of the intermediate density, velocity and internal energy (Eqs. (63)–(65)) to obtain a set of algebraic equations. However, the obtained equations are nonlinear, and the von Neumann stability analysis is only applicable to linear systems. In order to linearize the problem, the obtained equations are differentiated with respect to y_1, y_2, y_3 and y_4 to obtain the following system of equations

$$\mathrm{d}\mathbf{Y}^*=\left[\frac{\partial\mathbf{Y}^*}{\partial y_1}\right]^n\mathrm{d}y_1^n+\left[\frac{\partial\mathbf{Y}^*}{\partial y_2}\right]^n\mathrm{d}y_2^n+\left[\frac{\partial\mathbf{Y}^*}{\partial y_3}\right]^n\mathrm{d}y_3^n+\left[\frac{\partial\mathbf{Y}^*}{\partial y_4}\right]^n\mathrm{d}y_4^n \tag{75}$$

where $\mathbf{Y}^*=\left(y_1^*, y_2^*, y_3^*, y_4^*\right)^T$; and the superscript n represents the time level. Obviously, the terms inside the brackets are only related to the known functional values at the time level of n. Therefore, Eq. (75) is a linear system between $\mathrm{d}\mathbf{Y}^*$ and $\mathrm{d}y_1^n, \mathrm{d}y_2^n, \mathrm{d}y_3^n, \mathrm{d}y_4^n$, on which the von Neumann stability analysis can be applied. The following relationships are obtained:

$$\overline{\mathrm{d}y}_1^*=\overline{\mathrm{d}y}_1^n \quad \overline{\mathrm{d}y}_2^*=\overline{\mathrm{d}y}_2^n \quad \overline{\mathrm{d}y}_3^*=\overline{\mathrm{d}y}_3^n \quad \overline{\mathrm{d}y}_4^*=\overline{\mathrm{d}y}_4^n \tag{76}$$

where $\overline{\mathrm{d}y}_i^*$ and $\overline{\mathrm{d}y}_i^n$ are the magnitudes of $\mathrm{d}y_i^*=\overline{\mathrm{d}y}_i^* e^{i\mathbf{k}\cdot\mathbf{r}}$ and $\mathrm{d}y_i^n=\overline{\mathrm{d}y}_i^n e^{i\mathbf{k}\cdot\mathbf{r}}$ in the von Neumann stability analysis, respectively. The detailed derivation of Eq. (76) can be found in Appendix A.

The corrector step (Eqs. (53)–(55)) can also be analyzed through similar procedures, and finally the following relationships can be derived

$$\begin{cases} \overline{\mathrm{d}y}_1^{n+1}=\overline{\mathrm{d}y}_1^n \\ \overline{\mathrm{d}y}_2^{n+1}=\overline{\mathrm{d}y}_2^n \\ \overline{\mathrm{d}y}_3^{n+1}=g\beta T_m\delta_t\overline{\mathrm{d}y}_1^n+\overline{\mathrm{d}y}_3^n-g\beta\delta_t\overline{\mathrm{d}y}_4^n \\ \overline{\mathrm{d}y}_4^{n+1}=\overline{\mathrm{d}y}_4^n \end{cases} \tag{77}$$

The derivation process for Eq. (77) is presented in detail in Appendix A. Overall, the stability analysis of STLBM can be written in matrix form:

$$\overline{\mathrm{d}\mathbf{Y}}^{n+1}=\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ g\beta T_m\delta_t & 0 & 1 & -g\beta\delta_t \\ 0 & 0 & 0 & 1 \end{bmatrix}\cdot\overline{\mathrm{d}\mathbf{Y}}^n \tag{78}$$

It can be proven that all the eigenvalues in the characteristic matrix are 1. Therefore, the present simplified thermal lattice Boltzmann method is unconditionally stable.

3.3. Implementation of boundary conditions

In the present simplified thermal lattice Boltzmann method, the boundary values of the macroscopic variables are used in calculating the equilibrium distribution functions in the predictor step, while the boundary values of the non-equilibrium distribution functions are involved in the corrector step. Therefore, boundary conditions for both the macroscopic variables and the non-equilibrium distribution functions are required.

For the macroscopic variables, the implementation of the boundary conditions is identical to the conventional Navier–Stokes solvers. For instance, if it is a Dirichlet boundary condition, the boundary value of the variable is directly assigned on the boundary grid point. For a Neumann boundary condition, several inner layers of grid points are used to approximate the spatial gradient on the boundary. The number of layers involved depends on the order of accuracy to be achieved.

For the non-equilibrium distribution functions, the widely used non-equilibrium extrapolation scheme is applied in this paper. Specifically, the values of the non-equilibrium distribution functions on the boundary point are set to be equal to those of the grid point on the inner layer. Such treatment was initially proposed by Guo et al. [49], and has been proven to be of second-order accuracy in space [49].

3.4. Computational sequence

The computational sequence of the simplified thermal Lattice Boltzmann method can be summarized as follows:

(1) Specify the streaming distance $\delta_x(\delta_x=\delta_t)$. Determine the single relaxation parameters τ_v and τ_c.
(2) Predictor step. Use Eqs. (63)–(65) to calculate the intermediate physical properties ρ^*, $\mathbf{u}^*$ and e^*.
(3) Compute the non-equilibrium distribution functions (f_α^{neq} and g_α^{neq}) by applying Eqs. (69) and (70).
(4) Implement appropriate boundary conditions for the non-equilibrium distribution functions.
(5) Calculate the external force term $\mathbf{F}_E$ from Eq. (62).
(6) Corrector step. Obtain the density, velocity and internal energy at the new time step from Eqs. (66)–(68).
(7) Implement appropriate boundary conditions for the macroscopic physical properties.
(8) Repeat steps (2–7) until the computation converges or the prescribed maximum time step is reached.

4. Numerical examples

In this section, three representative numerical examples - the natural convection in a square cavity, the natural convection in a concentric annulus, and the mixed convection from a heated circular cylinder - are presented to demonstrate the robustness of the present simplified thermal LBM in simulating incompressible thermal viscous flows. For steady-state problems, the convergence criteria of the velocity and the temperature are:

$$\frac{\sum_{ij}\left|\left(\sqrt{u^2+v^2}\right)^{n+1}-\left(\sqrt{u^2+v^2}\right)^n\right|}{\sum_{ij}\left(\sqrt{u^2+v^2}\right)^{n+1}}\leqslant 10^{-8} \tag{79}$$

$$\frac{\sum_{ij}\left|T^{n+1}-T^n\right|}{\sum_{ij}\left|T^{n+1}\right|}\leqslant 10^{-8} \tag{80}$$

4.1. Natural convection in a square cavity

In the first example, we present the benchmark test of natural convection in a square cavity. As shown in Fig. 2, the computational domain is 1×1, bounded by stationary walls. The temperatures on the left and right walls are 1 and 0, respectively. Zero heat flux condition is implemented on the upper and lower walls. Physically, the temperature difference drives the bulk motion of the fluid until reaching a steady state.

For this problem, the flow pattern and the temperature distribution are determined by the Prandtl number and the Rayleigh number, defined as

$$\mathrm{Pr} = \frac{\upsilon}{\chi} \tag{81}$$

$$\mathrm{Ra} = \frac{g\beta \cdot \Delta T \cdot L^3}{\upsilon \cdot \chi} = \frac{V_c^2 \cdot L^2}{\upsilon \cdot \chi} \tag{82}$$

where L is the characteristic length of the square cavity; ΔT denotes the temperature difference between the left and right walls; and $V_c = \sqrt{g\beta L \cdot \Delta T}$ is the characteristic thermal velocity constrained by the low Mach number limit. Here, we set Pr = 0.71 and $V_c = 0.1$. For a given Rayleigh number, the kinematic viscosity and the thermal diffusivity can be determined from Eqs. (81) and (82); and the single relaxation parameters τ_υ and τ_c are obtained from the relationships given in Eqs. (27) and (28).

The Nusselt number Nu is an important parameter in thermal flows for evaluating the convective heat transfer rate. And the volume averaged Nusselt number $\overline{\mathrm{Nu}}$ over the whole flow domain is expressed as

$$\overline{\mathrm{Nu}} = \frac{L}{\chi \cdot \Delta T} \cdot \frac{1}{L^2} \int \int_\Omega \left(uT - \chi \frac{\partial T}{\partial x} \right) \mathrm{d}\Omega \tag{83}$$

where $q = uT - \chi \partial T/\partial x$ can be understood as the local heat flux in the horizontal direction. The local and average Nusslt numbers on the hot wall are respectively defined as

$$Nu = \frac{L}{\Delta T} \frac{\partial T}{\partial x} \tag{84}$$

$$Nu_0 = \frac{1}{L \cdot \Delta T} \int_{x=0} \frac{\partial T}{\partial x} \mathrm{d}y \tag{85}$$

The case of $\mathrm{Ra} = 10^4$ is selected for the convergence test. Numerical simulations were conducted on uniform grids with mesh sizes of 101×101, 151×151, 201×201, 251×251 and 301×301. The quantities and locations of the maximum vertical/horizontal velocities, as well as the average Nusselt number, are presented in Table 1. We can see that by refining the mesh, the flow properties are able to converge to certain values. A comparative study shows that the relative errors of the converged results to reference data given by Shu and Xue [50] are within 0.3%.

After the convergence test, we choose a case of high Rayleigh number ($\mathrm{Ra} = 10^6$) to numerically validate the stability of the present model. As shown in the von Neumann stability analysis in Section 3.2, the present STLBM is theoretically proven to be unconditionally stable. So in theory, the present method should be able to converge at any mesh size. We conducted 4 numerical tests with various mesh sizes (11×11, 51×51, 151×151 and 301×301), and found that all the tests were able to meet the convergence criteria. As can be seen from the converged isotherms presented in Fig. 3, the present method is able to give converged results even for very coarse meshes (e.g. 11×11), and the smoothness of the isotherms is well maintained. Such performance demonstrates that STLBM is a very stable method, and numerically validates the stability analysis conducted in Section 3.2.

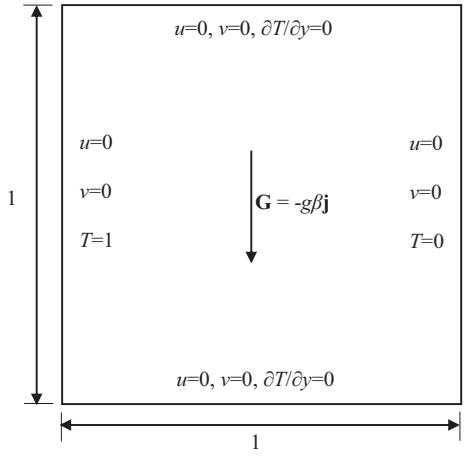


Fig. 2. Physical configuration of the setup for 2D natural convection in a square cavity.

In the following, 6 cases with a wide range of Rayleigh numbers from 10^3 to 10^8 are studied. It can be expected that, at higher Rayleigh number, the flow pattern is more complex with more abrupt changes. To numerically capture such abrupt changes, finer grids are required at higher Rayleigh numbers. In our simulations, uniform mesh sizes of 101×101, 151×151, 201×201, 301×301 are applied in cases with Rayleigh numbers of 10^3, 10^4, 10^5 and 10^6, respectively. And non-uniform meshes with mesh sizes of 301×301 and 401×401 are adopted in cases with Rayleigh numbers of 10^7 and 10^8, respectively.

Figs. 4 and 5 give the streamlines and the isothermal contours in our simulations respectively. In these figures, we observe that at lower Rayleigh number, the streamlines are regular circulation lines, and the isotherms have a generally parallel distribution from left to right. At high Rayleigh numbers, the flow changes become more severe. As a result, the velocity/temperature gradients increase significantly, which is reflected as streamlines and isotherms are squeezed more closely near the walls. The above observations are in good accordance with those reported in literature [33,51].

Converged values of hydrodynamic quantities and Nusselt numbers of our simulations are tabulated in Tables 2–4. Comprehensive comparisons with the data from references [17,21,33,37,38,50–53] are conducted in these tables. As can be seen, good agreement with the reference data is achieved in our simulations. Even at the highest Rayleigh number, i.e., $\mathrm{Ra} = 10^8$, the maximum relative error to the reference data is controlled within 0.5%. The above quantitative comparisons validate the accuracy of the present simplified thermal LBM.

4.2. Natural convection in a concentric annulus

In this section, to validate the performance of the simplified thermal LBM on non-uniform meshes and for problems with curved boundaries, we present a numerical test of the natural convection in a concentric annulus. The configuration of the problem is illustrated in Fig. 6. Basically, it is an internal natural convection problem bounded by a hot outer circular wall with temperature $T_o = 1$ and a cold inner circular wall with temperature $T_i = 0$. The aspect ratio is defined as the ratio of the outer to inner radius, i.e., $\mathrm{Ar} = R_o/R_i$. The characteristic length L, which is used to define the Rayleigh number (Eq. (82)), is the radial distance between the two circular walls, i.e., $L = R_o - R_i$. In the present study, we choose Ar = 2.6 and Pr = 0.71. Six cases with various Rayleigh numbers of $\mathrm{Ra} = 10^2$, 10^3, 3×10^3, 6×10^3, 10^4 and 5×10^4 are carried out on a body-fitted O-type mesh size of 251×61. It should be noted that due to the aforementioned lattice uniformity, the present SLBM cannot be directly implemented on non-uniform meshes. Therefore, in the present simulations, Lagrange interpolation algorithm is introduced to determine physical properties at locations where the streaming process initiates, with which we can ensure that the collision process happens on the central grid point. To quantify the heat transfer efficiency on the surfaces, the equivalent heat conductivities are defined as

$$\overline{k_{eqi}} = \frac{\ln(\mathrm{Ar})}{2\pi(\mathrm{Ar}-1)} \int_0^{2\pi} \frac{\partial T}{\partial r} \mathrm{d}\theta, \quad \text{for the inner surface} \tag{86}$$

Table 1
Convergence study of the natural convection in a square cavity at Ra = 10^4.

Method	Mesh	u_{max}	y	v_{max}	x	$\overline{Nu}$
Present	101 × 101	16.112	0.820	19.551	0.120	2.238
	151 × 151	16.137	0.820	19.594	0.120	2.243
	201 × 201	16.152	0.825	19.611	0.120	2.245
	251 × 251	16.160	0.824	19.619	0.120	2.245
	301 × 301	16.165	0.823	19.624	0.120	2.245
DQ [50]		16.190	0.825	19.638	0.120	2.245
Relative error		0.15%	0.24%	0.07%	0%	0%

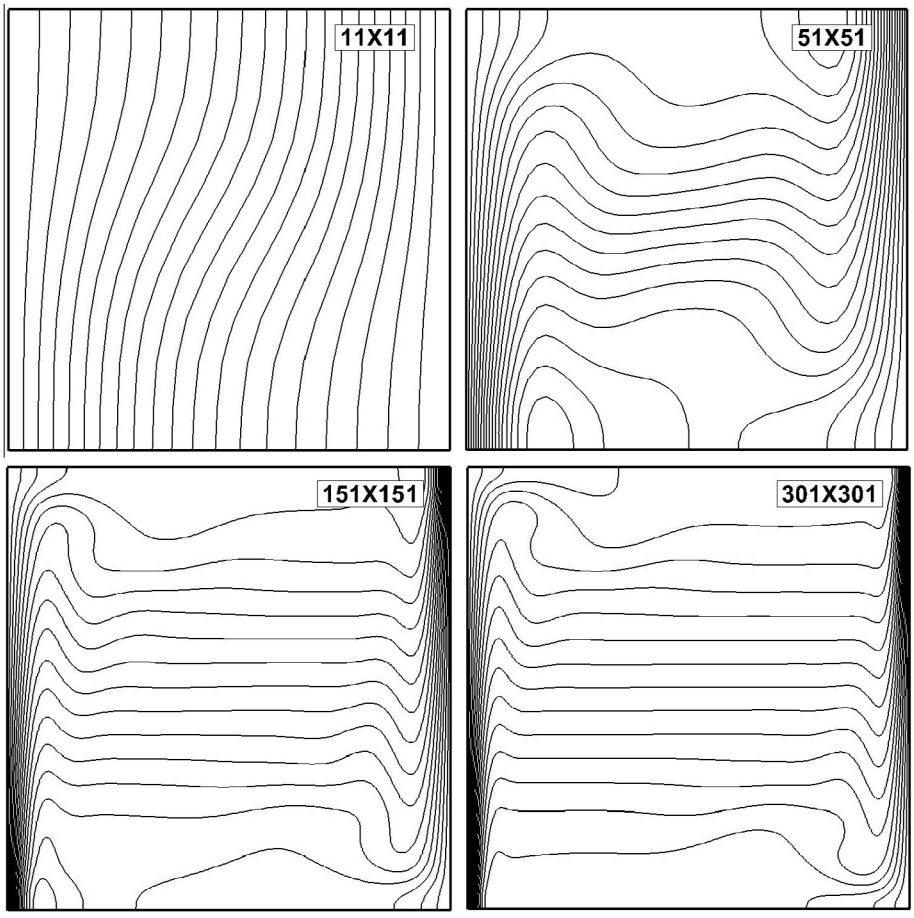


Fig. 3. Isotherms for different mesh sizes at Ra = 10^6.

$$\overline{k_{eqo}} = \frac{\mathrm{Ar}\cdot\ln(\mathrm{Ar})}{2\pi(\mathrm{Ar}-1)}\int_0^{2\pi}\frac{\partial T}{\partial r}\mathrm{d}\theta, \quad \text{for the outer surface} \tag{87}$$

The converged results of the above equivalent conductivities in our simulations are compared with the data given by Shu [54] and Wang et al. [37] in Table 5. We observe that the equivalent conductivities increase with increasing Rayleigh number, indicating higher heat transfer efficiency. We also observe that the computed parameters agree well with the reference data. Even in the case with the highest Rayleigh number, the maximum relative error is within 1.5%.

Fig. 7 presents the isotherms at different Rayleigh numbers in our simulations. Similar to the observations for natural convection in a square cavity, at higher Rayleigh number, the isotherms are squeezed closer to the boundaries. Such phenomenon corresponds with higher conductivities on the walls at larger Rayleigh numbers, as presented in the quantitative comparisons above.

4.3. Mixed heat transfer from a heated circular cylinder

The previous two examples exhibit the robustness of the simplified thermal lattice Boltzmann method in simulating internal thermal flows. In this section, mixed heat transfer from a heated circular cylinder is simulated to test the performance of the present method in modeling external thermal flows. "Mixed heat transfer" means that the problem involves both natural heat convection and forced heat convection. Therefore, the physical mechanism is relatively more complex. In the present example (see Fig. 8), a stationary circular cylinder is placed in an incoming fluid flow with free-stream velocity $U_\infty = 0.1$ and free-stream temperature $T_\infty = 0$. The circular cylinder has diameter $D = 1$, and its surface is at a constant temperature $T_s = 1.0$. Apart from the Prandtl number, here we introduce several other important non-dimensional parameters: Reynolds number, Grashof number,

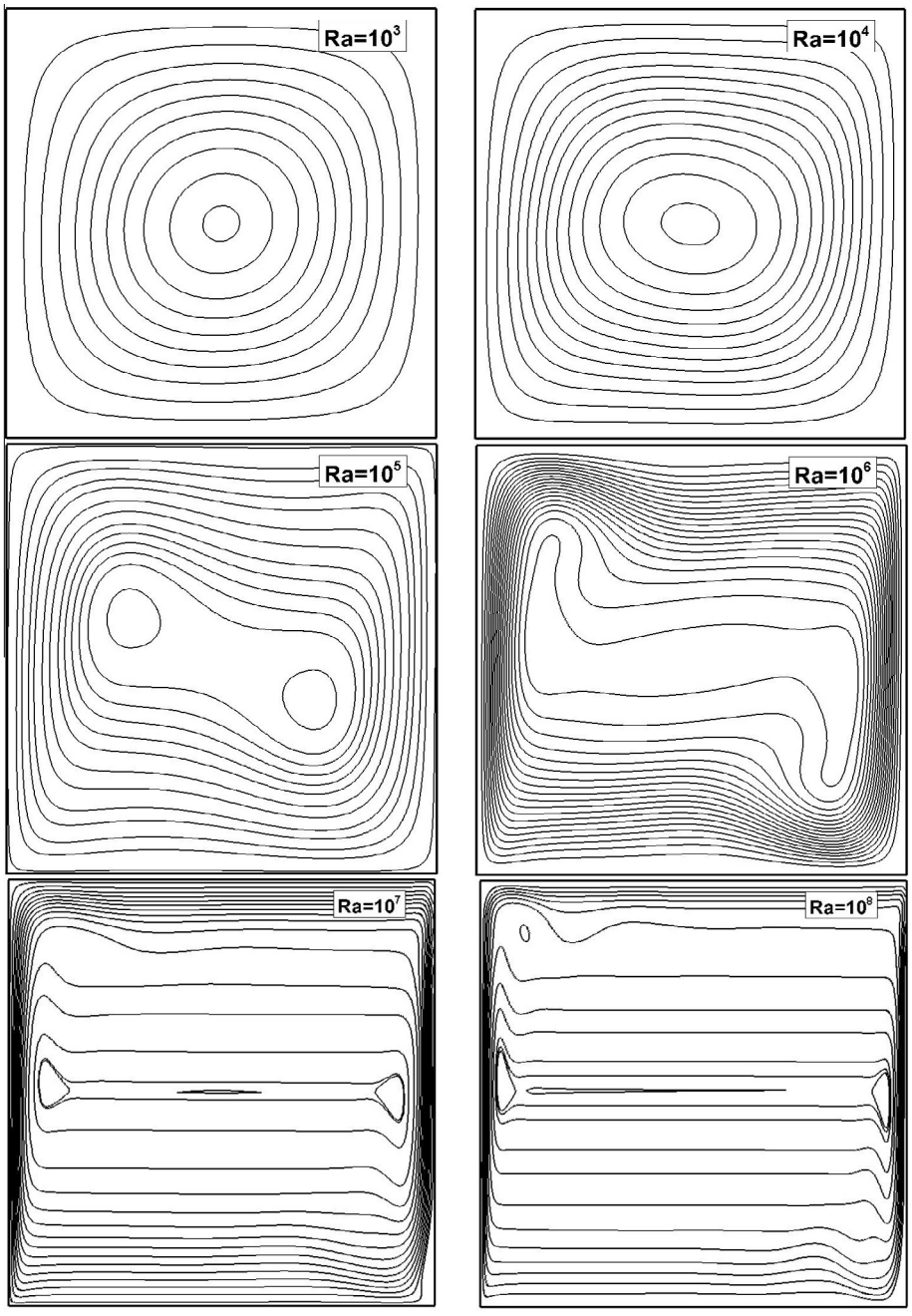


Fig. 4. Streamlines at different Rayleigh numbers.

Richardson number, average Nusselt number and Strouhal number, which are respectively defined as

$$Re = \frac{U_\infty D}{\upsilon} \tag{88}$$

$$\mathrm{Gr} = \frac{g\beta(T_s - T_\infty)D^3}{\upsilon^2} \tag{89}$$

$$\mathrm{Ri} = \frac{\mathrm{Gr}}{Re^2} \tag{90}$$

$$\overline{\mathrm{Nu}} = -\frac{D}{2\pi(T_s - T_\infty)} \int_0^{2\pi} \left.\frac{\partial T}{\partial n}\right|_s \mathrm{d}\theta \tag{91}$$

$$\mathrm{St} = \frac{fD}{U_\infty} \tag{92}$$

where f is the oscillating frequency in unsteady cases at higher Reynolds number. A typical O-type structured grid is applied in our simulation; the mesh sizes are 301×201 and 301×501 for the steady and unsteady cases, respectively. The reference density of the fluid is set to $\rho_\infty = 1$, while the Prandtl number is chosen as $\mathrm{Pr} = 0.7$ to keep in accordance with the initial set-up in references [37,55].

The mixed heat transfer problem is affected by both the forced external flow and the temperature field. Therefore, its flow shape is determined by both the Reynolds number and the Grashof number. At low Reynolds number, e.g., $\mathrm{Re} = 20$ in the present example, the

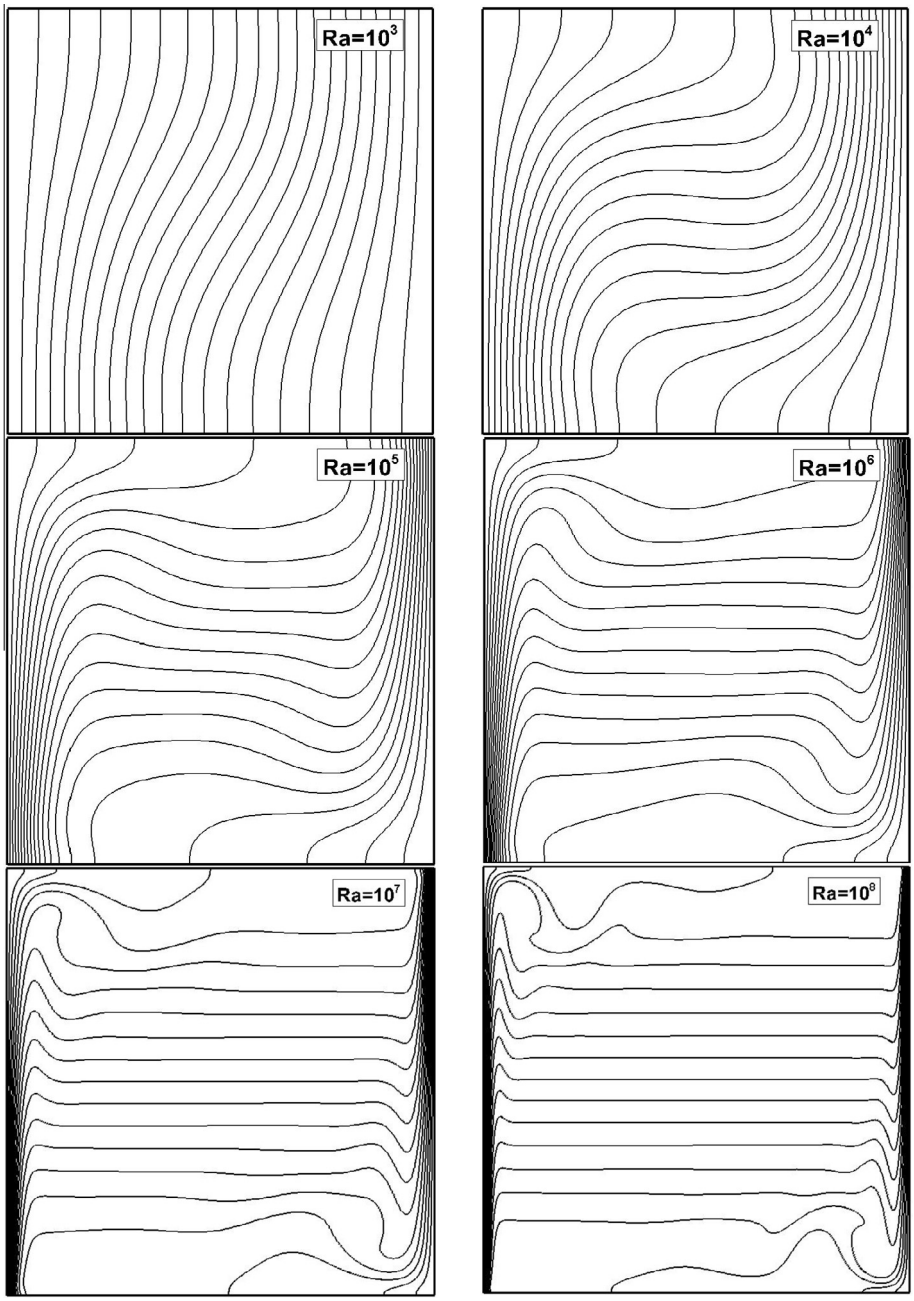


Fig. 5. Isotherms at different Rayleigh numbers.

flow is considered to be steady. Figs. 9 and 10 give the streamlines and isotherms of different Grashof number at Re = 20, respectively. We observe that, at zero Grashof number, the temperature has no effect on the velocity field. Therefore, clear recirculation zones are captured and the streamlines obtained are the same as those in isothermal simulations at the same Reynolds number. When Grashof number increases, the temperature and velocity fields are coupled in the computations and the viscous effect is then weakened by the heat convection. In the streamline pictures, we observe that the recirculation zones keep shrinking with increasing Grashof number, until they fully disappear after $Gr \geqslant 800$. Such phenomenon means that at higher Grashof number, heat convection plays the dominant role in the flow pattern. In the isotherm figures, we see that at higher Grashof number, the isotherms are squeezed closer to the cylinder surface. Therefore high heat transfer rates could be expected in such cases. For a quantitative comparison, we present the average Nusselt number on the cylinder and the separation angle at different Grashof numbers in Table 6. As expected, at higher Grashof number, a larger average Nusselt number is obtained in our simulations. These quantities agree well with reported literature [37,55].

Now, we consider the cases at Re = 100. For isothermal flows past a circular cylinder, the flow is unsteady at this Reynolds number. Here, 3 cases with Richardson numbers of Ri = 0, 0.1 and 0.5 are simulated by the present STLBM. The streamlines and the isotherms are shown in Figs. 11 and 12, respectively. We observe from Fig. 11 that at low Richardson number, the flows are still unsteady, with the occurrence of periodic vortex shedding. In the case with Ri = 0.5, however, the flow becomes steady with stable recirculation zones after the cylinder. This flow phenomenon is in good agreement with previous studies [37,55,56]. For quantitative comparisons, we tabulate the average Nusselt number on the surface of

Table 2
Results of natural convection at different Rayleigh numbers.

Ra		10^3	10^4	10^5	10^6
u_{max}	Present	3.648	16.137	34.566	64.173
	TLBM [33]	3.644	16.134	34.261	63.024
	TLBM [21]	3.653	16.237	34.823	64.870
	TLBFS [37]	3.640	16.14	34.87	64.838
	DQ [50]	3.649	16.190	34.736	64.775
	Davis [51]	3.634	16.182	34.81	65.33
y	Present	0.810	0.820	0.855	0.850
	TLBM [33]	0.810	0.820	0.855	0.848
	TLBM [21]	0.813	0.820	0.853	0.853
	TLBFS [37]	0.815	0.825	0.855	0.850
	DQ [50]	0.815	0.825	0.855	0.850
	Davis [51]	0.813	0.823	0.855	0.851
v_{max}	Present	3.699	19.594	68.351	218.56
	TLBM [33]	3.691	19.552	67.799	215.26
	TLBM [21]	3.701	19.680	68.712	221.19
	TLBFS [37]	3.708	19.67	68.85	220.92
	DQ [50]	3.698	19.638	68.640	220.64
	Davis [51]	3.679	19.509	68.22	216.75
x	Present	0.180	0.120	0.065	0.040
	TLBM [33]	0.180	0.120	0.065	0.040
	TLBM [21]	0.180	0.117	0.064	0.039
	TLBFS [37]	0.180	0.118	0.065	0.038
	DQ [50]	0.180	0.120	0.065	0.035
	Davis [51]	0.179	0.120	0.066	0.039
$\overline{Nu}$	Present	1.118	2.243	4.498	8.713
	TLBM [33]	1.117	2.241	4.511	8.731
	TLBM [21]	-	-	-	-
	TLBFS [37]	1.115	2.232	4.491	8.711
	DQ [50]	1.118	2.245	4.523	8.762
	Davis [51]	1.116	2.234	4.510	8.798

Table 3
Comparisons of Nusselt numbers at the hot wall.

Ra		10^3	10^4	10^5	10^6
Nu_0	Present	1.118	2.249	4.526	8.732
	TLBM [21]	1.117	2.235	4.504	8.767
	Davis [51]	1.116	2.242	4.532	8.928
Nu_{max}	Present	1.508	3.522	7.804	17.396
	TLBM [21]	1.501	3.507	7.658	17.288
	Davis [51]	1.501	3.545	7.761	18.076
y	Present	0.090	0.140	0.075	0.0367
	TLBM [21]	0.086	0.148	0.088	0.0441
	Davis [51]	0.087	0.149	0.085	0.0456
Nu_{min}	Present	0.691	0.591	0.753	1.050
	TLBM [21]	0.698	0.584	0.728	0.998
	Davis [51]	0.694	0.592	0.736	1.005
y	Present	0.980	1.000	1.000	0.993
	TLBM [21]	0.953	0.984	0.990	0.990
	Davis [51]	1.000	1.000	1.000	1.000

the cylinder and the Strouhal number at Ri = 0 in Table 7. The numerical results obtained in our simulations agree well with the reported results in literature [37,57–60].

5. Concluding remarks

In this paper, a simplified thermal lattice Boltzmann method (STLBM) is developed for simulating incompressible thermal flows. With the assistance of fractional step technique, the macroscopic governing equations recovered from thermal lattice Boltzmann method through Chapman–Enskog expansion analysis can be solved using a predictor–corrector scheme. In STLBM, the formulations in the predictor–corrector steps are reconstructed from lattice properties and the relationships given by C–E analysis, which only involve the equilibrium and the non-equilibrium distribution functions. In the computations, the equilibrium distribution functions are calculated from the macroscopic variables, while the non-equilibrium distribution functions are simply evaluated from the differences of two equilibrium distribution functions at different locations and time levels. As a result, STLBM directly updates the macroscopic flow variables rather than the distribution functions, which greatly saves the virtual memories. At the same time, the physical boundary conditions, which are related to the macroscopic variables, can be directly implemented in STLBM. We prove this method to be unconditionally stable using the von Neumann stability analysis, and validate it numerically using a series of natural convection systems at high Rayleigh number.

Three typical numerical examples, namely, natural convection in a square cavity, natural convection in a concentric annulus

Table 4
Comparisons of representative quantities at Ra = 10^7 and 10^8.

Ra		10^7	10^8
Nu_0	Present	16.510	30.12
	Contrino et al. [53]	16.523	30.22
	Quere [17]	16.523	30.23
	Mayne et al. [52]	16.387	29.63
	Wang et al. [38]	–	–
$Nu_{1/2}$	Present	16.485	30.081
	Contrino et al. [53]	16.523	30.227
	Quere [17]	16.52	30.225
	Mayne et al. [52]	–	–
	Wang et al. [38]	16.543	30.301
u_{max}	Present	146.78	306.67
	Contrino et al. [53]	148.58	321.37
	Quere [17]	148.59	321.9
	Mayne et al. [52]	145.27	283.07
	Wang et al. [38]	148.84	323.65
y	Present	0.879	0.927
	Contrino et al. [53]	0.8793	0.9276
	Quere [17]	0.879	0.928
	Mayne et al. [52]	0.8845	0.9455
	Wang et al. [38]	0.8789	0.9288
v_{max}	Present	699.56	2222.1
	Contrino et al. [53]	699.31	2222.3
	Quere [17]	699.18	2222
	Mayne et al. [52]	703.25	2223.4
	Wang et al. [38]	699.91	2222.9
x	Present	0.021	0.012
	Contrino et al. [53]	0.0213	0.0120
	Quere [17]	0.021	0.012
	Mayne et al. [52]	0.0215	0.0130
	Wang et al. [38]	0.0216	0.1192

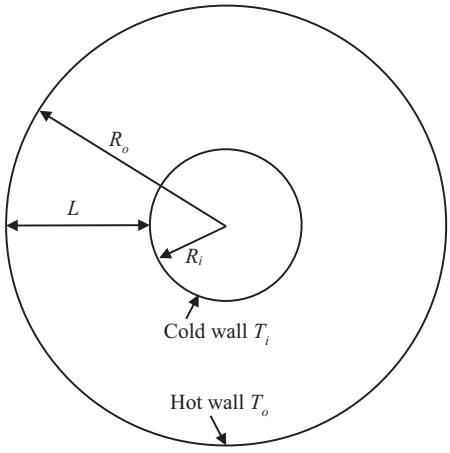


Fig. 6. Illustration of the setup for natural convection in a concentric annulus.

and mixed heat transfer from a heated circular cylinder, are presented to evaluate the performance of the present method in simulating various kinds of thermal flows, as well as its flexibility in problems with curved boundaries and on non-uniform meshes. Good agreement with the reported data from literature is observed for the above simulations, which exhibits the robustness of the simplified thermal lattice Boltzmann method proposed in this paper.

Appendix A. Von Neumann stability analysis of the simplified thermal lattice Boltzmann method

In this section, we perform von Neumann stability analysis on the simplified thermal lattice Boltzmann method. With the notation given in Eq. (71), the density distribution function and the internal energy distribution function can be expressed respectively as Eqs. (73) and (74). Then the intermediate density, velocity and internal energy (Eqs. (63)–(65)) can be written as

$$y_1^* = \sum_{\alpha=0}^{8}\left[w_\alpha y_{1\alpha}^n + \frac{w_\alpha}{c_s^2}\left(e_{\alpha x}y_{2\alpha}^n + e_{\alpha y}y_{3\alpha}^n\right) + \frac{w_\alpha}{2c_s^4}\frac{\left(e_{\alpha x}y_{2\alpha}^n + e_{\alpha y}y_{3\alpha}^n\right)^2}{y_{1\alpha}^n} - \frac{w_\alpha}{2c_s^2}\frac{\left(y_{2\alpha}^n\right)^2 + \left(y_{3\alpha}^n\right)^2}{y_{1\alpha}^n}\right] \tag{A-1}$$

$$y_2^* = \sum_{\alpha=0}^{8}\left[w_\alpha e_{\alpha x}y_{1\alpha}^n + \frac{w_\alpha e_{\alpha x}}{c_s^2}\left(e_{\alpha x}y_{2\alpha}^n + e_{\alpha y}y_{3\alpha}^n\right) + \frac{w_\alpha e_{\alpha x}}{2c_s^4}\frac{\left(e_{\alpha x}y_{2\alpha}^n + e_{\alpha y}y_{3\alpha}^n\right)^2}{y_{1\alpha}^n} - \frac{w_\alpha e_{\alpha x}}{2c_s^2}\frac{\left(y_{2\alpha}^n\right)^2 + \left(y_{3\alpha}^n\right)^2}{y_{1\alpha}^n}\right] \tag{A-2}$$

$$y_3^* = \sum_{\alpha=0}^{8}\left[w_\alpha e_{\alpha y}y_{1\alpha}^n + \frac{w_\alpha e_{\alpha y}}{c_s^2}\left(e_{\alpha x}y_{2\alpha}^n + e_{\alpha y}y_{3\alpha}^n\right) + \frac{w_\alpha e_{\alpha y}}{2c_s^4}\frac{\left(e_{\alpha x}y_{2\alpha}^n + e_{\alpha y}y_{3\alpha}^n\right)^2}{y_{1\alpha}^n} - \frac{w_\alpha e_{\alpha y}}{2c_s^2}\frac{\left(y_{2\alpha}^n\right)^2 + \left(y_{3\alpha}^n\right)^2}{y_{1\alpha}^n}\right] \tag{A-3}$$

$$\begin{aligned} y_4^* = {} & \sum_{\alpha=1}^{8}\frac{1}{6}y_{4\alpha}^n\frac{e_{\alpha x}y_{2\alpha}^n + e_{\alpha y}y_{3\alpha}^n}{y_{1\alpha}^n} \\ & + \sum_{\alpha=1}^{4}\left[\frac{1}{6}y_{4\alpha}^n + \frac{1}{2}y_{4\alpha}^n\left(\frac{e_{\alpha x}y_{2\alpha}^n + e_{\alpha y}y_{3\alpha}^n}{y_{1\alpha}^n}\right)^2 - \frac{1}{6}y_{4\alpha}^n\frac{\left(y_{2\alpha}^n\right)^2 + \left(y_{3\alpha}^n\right)^2}{\left(y_{1\alpha}^n\right)^2}\right] \\ & + \sum_{\alpha=5}^{8}\left[\frac{1}{12}y_{4\alpha}^n + \frac{1}{8}y_{4\alpha}^n\left(\frac{e_{\alpha x}y_{2\alpha}^n + e_{\alpha y}y_{3\alpha}^n}{y_{1\alpha}^n}\right)^2 - \frac{1}{24}y_{4\alpha}^n\frac{\left(y_{2\alpha}^n\right)^2 + \left(y_{3\alpha}^n\right)^2}{\left(y_{1\alpha}^n\right)^2}\right] \\ & - \frac{2}{3}y_{40}^n\frac{\left(y_{20}^n\right)^2 + \left(y_{30}^n\right)^2}{\left(y_{10}^n\right)^2} \end{aligned} \tag{A-4}$$

where the superscript n represents the time step. To linearize the problem, Eqs. (A-1)–(A-4) are differentiated with respect to y_1, y_2, y_3 and y_4

Table 5
Comparison of average equivalent heat conductivity at different Rayleigh numbers.

Ra	Inner cylinder, $\overline{k_{eqi}}$			Outer cylinder, $\overline{k_{eqo}}$		
	Wang et al. [37]	Shu [54]	Present	Wang et al. [37]	Shu [54]	Present
10^2	1.002	1.001	0.986	1.002	1.001	1.007
10^3	1.076	1.082	1.091	1.078	1.082	1.082
3×10^3	1.381	1.397	1.393	1.384	1.397	1.403
6×10^3	1.695	1.715	1.698	1.701	1.715	1.727
10^4	1.960	1.979	1.947	1.960	1.979	1.994
5×10^4	2.941	2.958	2.909	2.941	2.958	2.975

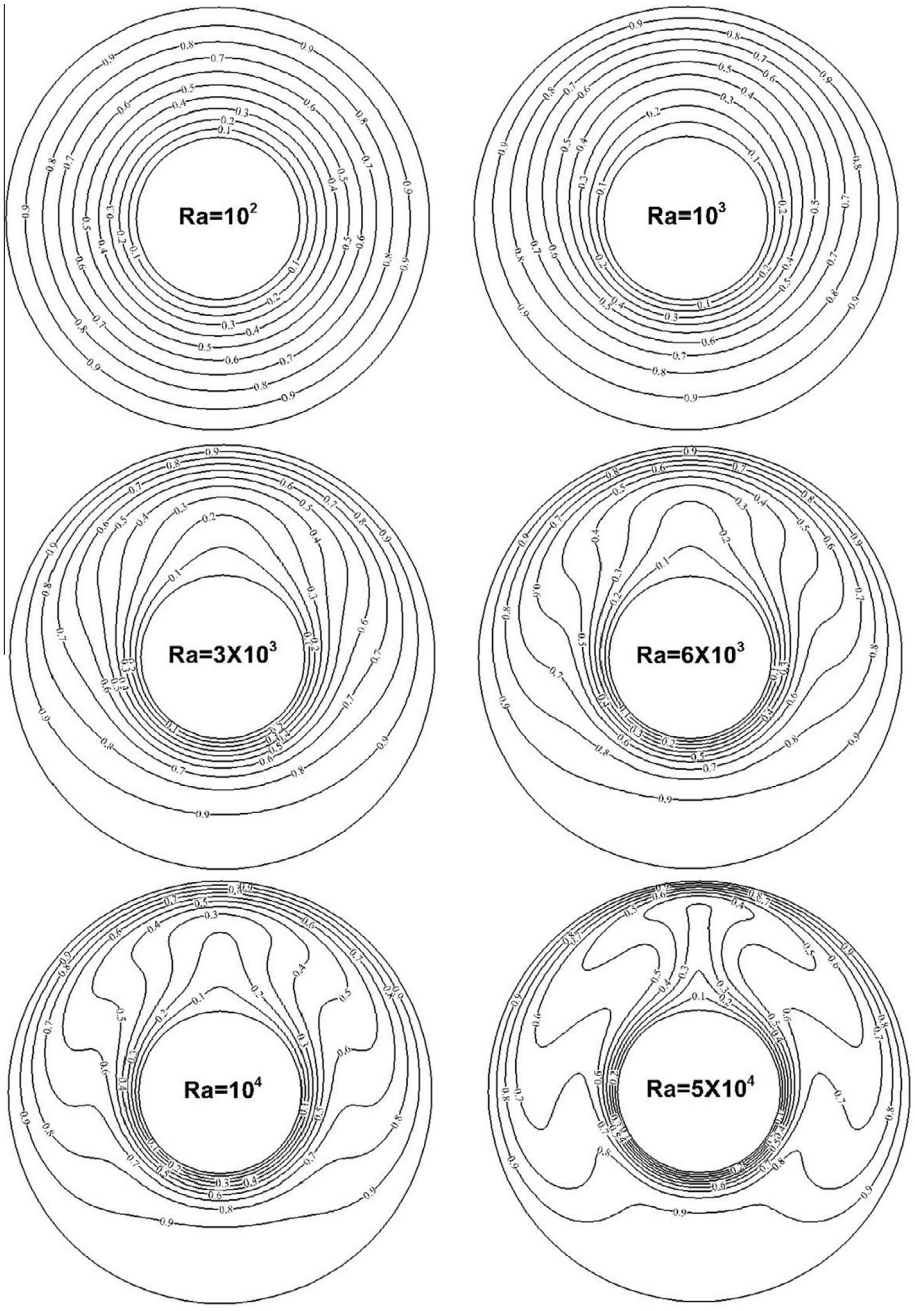


Fig. 7. Isotherms at different Rayleigh numbers.

$$dy_1^* = \left(\frac{\partial y_1^*}{\partial y_1}\right)^n dy_1^n + \left(\frac{\partial y_1^*}{\partial y_2}\right)^n dy_2^n + \left(\frac{\partial y_1^*}{\partial y_3}\right)^n dy_3^n \tag{A-5}$$

$$dy_2^* = \left(\frac{\partial y_2^*}{\partial y_1}\right)^n dy_1^n + \left(\frac{\partial y_2^*}{\partial y_2}\right)^n dy_2^n + \left(\frac{\partial y_2^*}{\partial y_3}\right)^n dy_3^n \tag{A-6}$$

$$dy_3^* = \left(\frac{\partial y_3^*}{\partial y_1}\right)^n dy_1^n + \left(\frac{\partial y_3^*}{\partial y_2}\right)^n dy_2^n + \left(\frac{\partial y_3^*}{\partial y_3}\right) dy_3^n \tag{A-7}$$

$$dy_4^* = \left(\frac{\partial y_4^*}{\partial y_1}\right)^n dy_1^n + \left(\frac{\partial y_4^*}{\partial y_2}\right)^n dy_2^n + \left(\frac{\partial y_4^*}{\partial y_3}\right)^n dy_3^n + \left(\frac{\partial y_4^*}{\partial y_4}\right)^n dy_4^n \tag{A-8}$$

For illustration purposes, the first term in Eq. (A-8) is

$$\begin{aligned}\left(\frac{\partial y_4^*}{\partial y_1}\right)^n dy_1^n = \Bigg\{ & -\sum_{\alpha=1}^{8} \frac{y_{4\alpha}^n}{6\left(y_{1\alpha}^n\right)^2}\left(e_{\alpha x} y_{2\alpha}^n + e_{\alpha y} y_{3\alpha}^n\right) \\ & -\sum_{\alpha=1}^{4} \frac{y_{4\alpha}^n}{\left(y_{1\alpha}^n\right)^3}\left[\left(e_{\alpha x} y_{2\alpha}^n + e_{\alpha y} y_{3\alpha}^n\right)^2 - \frac{1}{3}\left(\left(y_{2\alpha}^n\right)^2 + \left(y_{3\alpha}^n\right)^2\right)\right] \\ & -\sum_{\alpha=5}^{8} \frac{y_{4\alpha}^n}{\left(y_{1\alpha}^n\right)^3}\left[\frac{1}{4}\left(e_{\alpha x} y_{2\alpha}^n + e_{\alpha y} y_{3\alpha}^n\right)^2 - \frac{1}{12}\left(\left(y_{2\alpha}^n\right)^2 + \left(y_{3\alpha}^n\right)^2\right)\right] \\ & + \frac{4}{3}\frac{y_{40}^n}{\left(y_{10}^n\right)^3}\left(\left(y_{20}^n\right)^2 + \left(y_{30}^n\right)^2\right)\Bigg\} dy_1^n \end{aligned} \tag{A-9}$$

Following the von-Neumann stability analysis [48], these variables are approximated by a Fourier series expansion, of which we consider a random component of these variables, which can be written as

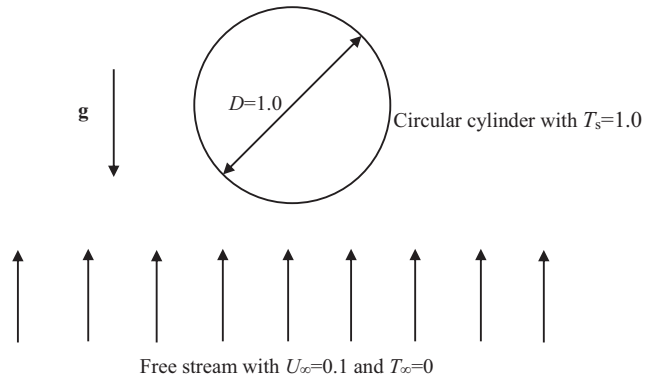


Fig. 8. Illustration of the mixed heat transfer simulation setup from a heated circular cylinder.

$$dy_i^* = \overline{dy}_i^* e^{i\mathbf{k}\cdot\mathbf{r}}; \quad dy_i^n = \overline{dy}_i^n e^{i\mathbf{k}\cdot\mathbf{r}}; \quad y_{i\alpha}^n = y_i^n e^{i\mathbf{k}\cdot(\mathbf{r}-\mathbf{e}_\alpha\delta_t)}, \ i = 1,2,3,4 \tag{A-10}$$

For instance, after substituting Eq. (A-10), the first term in Eq. (A-9) can be expressed as:

$$\begin{aligned}
&-\sum_{\alpha=1}^{8} \frac{y_{4\alpha}^n}{6\left(y_{1\alpha}^n\right)^2}\left(e_{\alpha x}y_{2\alpha}^n + e_{\alpha y}y_{3\alpha}^n\right) \\
&= -\sum_{\alpha=1}^{8} \frac{y_4^n e^{i\mathbf{k}\cdot(\mathbf{r}-\mathbf{e}_\alpha\delta_t)}}{6\left(y_1^n\right)^2 e^{2i\mathbf{k}\cdot(\mathbf{r}-\mathbf{e}_\alpha\delta_t)}}\left(e_{\alpha x}y_2^n e^{i\mathbf{k}\cdot(\mathbf{r}-\mathbf{e}_\alpha\delta_t)} + e_{\alpha y}y_3^n e^{i\mathbf{k}\cdot(\mathbf{r}-\mathbf{e}_\alpha\delta_t)}\right) \\
&= -\frac{y_4^n y_2^n}{6\left(y_1^n\right)^2}\sum_{\alpha=1}^{8} e_{\alpha x} - \frac{y_4^n y_3^n}{6\left(y_1^n\right)^2}\sum_{\alpha=1}^{8} e_{\alpha y} = 0
\end{aligned} \tag{A-11}$$

Through similar manipulations of the remaining terms, Eq. (A-9) is reduced to

$$\begin{aligned}
\left(\frac{\partial y_4^*}{\partial y_1}\right)^n dy_1^n = \overline{dy}_1^n e^{i\mathbf{k}\cdot\mathbf{r}}\Bigg\{ &-\sum_{\alpha=1}^{4}\frac{y_4^n}{\left(y_1^n\right)^3}\left[\left(e_{\alpha x}y_2^n + e_{\alpha y}y_3^n\right)^2 - \frac{1}{3}\left(\left(y_2^n\right)^2 + \left(y_3^n\right)^2\right)\right] \\
&-\sum_{\alpha=5}^{8}\frac{y_4^n}{\left(y_1^n\right)^3}\left[\frac{1}{4}\left(e_{\alpha x}y_2^n + e_{\alpha y}y_3^n\right)^2 - \frac{1}{12}\left(\left(y_2^n\right)^2 + \left(y_3^n\right)^2\right)\right] \\
&+\frac{4}{3}\frac{y_4^n}{\left(y_1^n\right)^3}\left(\left(y_2^n\right)^2 + \left(y_3^n\right)^2\right)\Bigg\} \\
= \overline{dy}_1^n e^{i\mathbf{k}\cdot\mathbf{r}}&\left(-2 + \frac{4}{3} - 1 + \frac{1}{3} + \frac{4}{3}\right)\frac{y_4^n}{\left(y_1^n\right)^3}\left(\left(y_2^n\right)^2 + \left(y_3^n\right)^2\right) = 0
\end{aligned} \tag{A-12}$$

Note that the following isentropic properties of lattice tensors for D2Q9 model have been used in simplifying Eqs. (A-11) and (A-12)

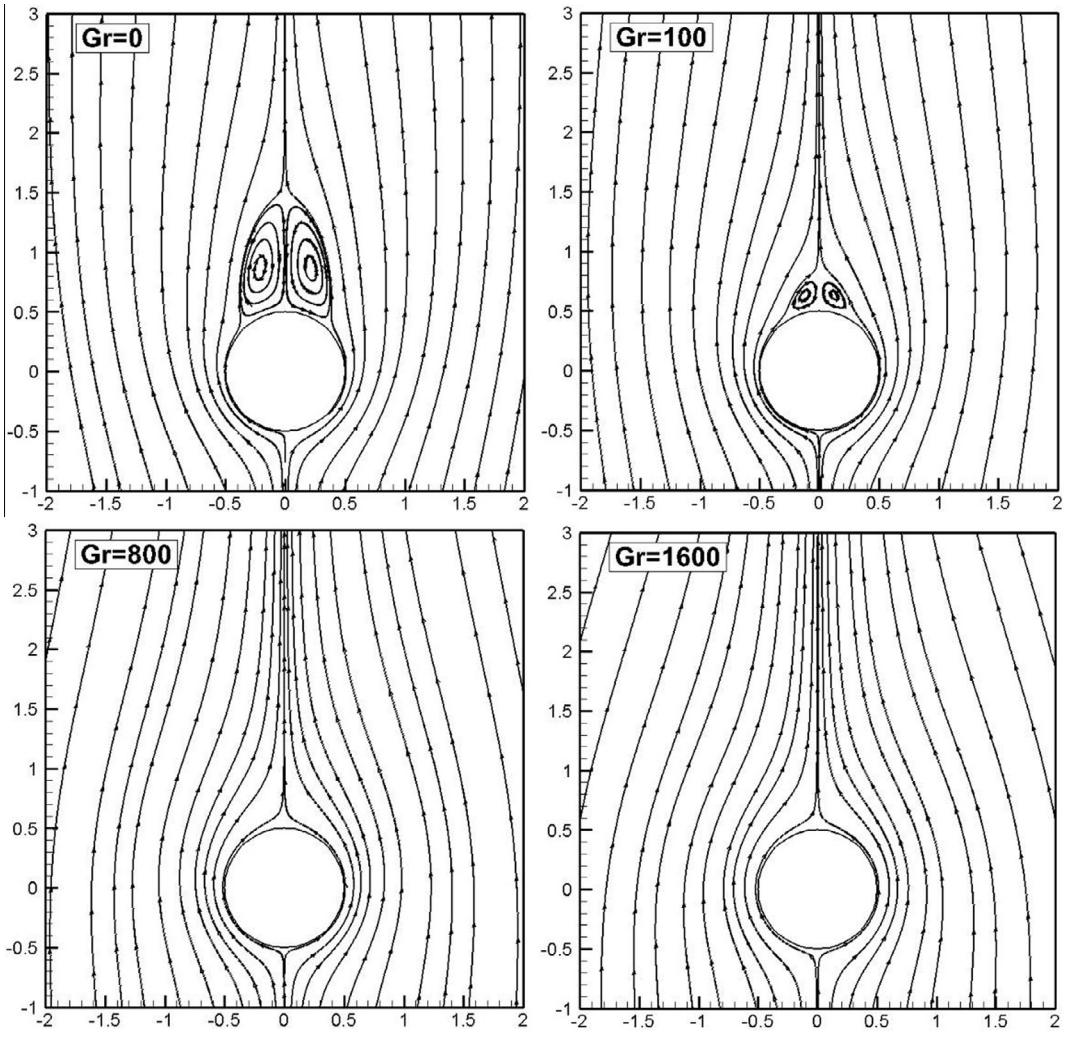


Fig. 9. Streamlines of mixed heat transfer system at Re = 20 for different Grashof numbers.

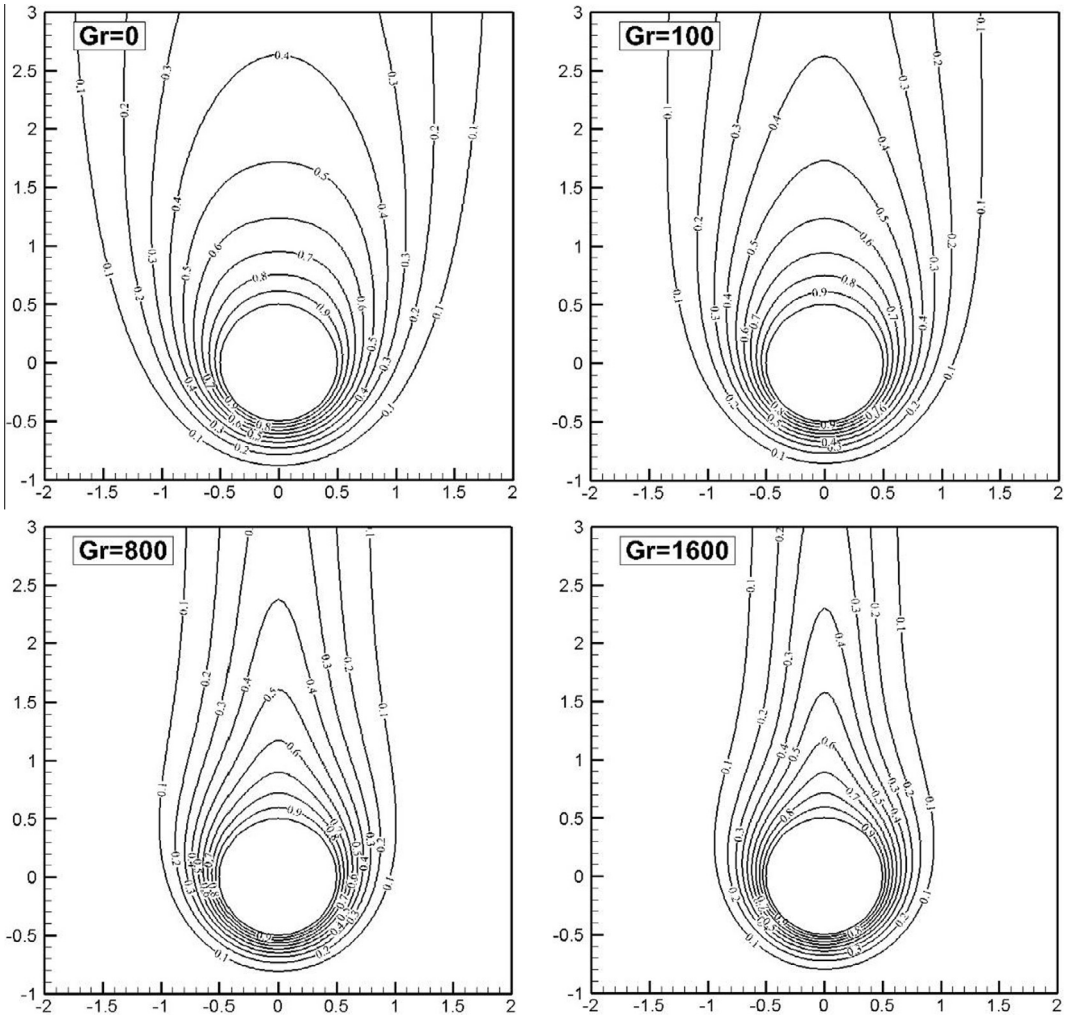


Fig. 10. Isotherms of mixed heat transfer system at Re = 20 for different Grashof numbers.

Table 6
Comparison of average Nusselt number and separation angle for mixed heat convection at Re = 20.

	Gr	$\overline{Nu}$			θ_s (deg)		
		Present	Wang et al. [37]	Badr [55]	Present	Wang et al. [37]	Badr [55]
Re = 20	0	2.468	2.523	2.540	43.78	43.19	43.13
	100	2.652	2.654	2.640	29.54	29.73	29.51
	800	3.204	3.208	3.227	0.0	0.0	0.0
	1600	3.511	3.554	3.564	0.0	0.0	0.0

$$\sum_\alpha w_\alpha e_{\alpha x}^2 e_{\alpha y} = \sum_\alpha w_\alpha e_{\alpha x} e_{\alpha y}^2 = \sum_\alpha w_\alpha e_{\alpha x}^3 = \sum_\alpha w_\alpha e_{\alpha y}^3 = 0$$
$$\sum_\alpha w_\alpha e_{\alpha x}^2 = \sum_\alpha w_\alpha e_{\alpha y}^2 = c_s^2$$
$$\sum_\alpha w_\alpha e_{\alpha x} e_{\alpha y} = 0 \qquad \text{(A-13)}$$
$$\sum_\alpha w_\alpha = 1$$
$$\sum_\alpha w_\alpha e_{\alpha x} = \sum_\alpha w_\alpha e_{\alpha y} = 0$$

By carrying out similar analysis for the remaining terms, Eqs. (A-5)–(A-8) can be derived as

$$\overline{dy}_1^* = \overline{dy}_1^n \quad \overline{dy}_2^* = \overline{dy}_2^n \quad \overline{dy}_3^* = \overline{dy}_3^n \quad \overline{dy}_4^* = \overline{dy}_4^n \qquad \text{(A-14)}$$

In the corrector step (Eqs. (53)–(55)), the density, the velocity and the energy in the next time step are evaluated by

$$y_1^{n+1} = y_1^* \qquad \text{(A-15)}$$

$$\begin{aligned} y_2^{n+1} &= y_2^* + \left(1 - \tfrac{1}{\tau_\upsilon}\right) \sum_\alpha e_{\alpha x} f_\alpha^{neq}(\mathbf{r} - \mathbf{e}_\alpha \delta_t, (n+1)\Delta t) \\ &= y_2^* - (\tau_\upsilon - 1) \sum_\alpha e_{\alpha x} \left[f_\alpha^{eq}(\mathbf{r} - \mathbf{e}_\alpha \delta_t, (n+1)\Delta t) - f_\alpha^{eq}(\mathbf{r} - 2\mathbf{e}_\alpha \delta_t, n\Delta t) \right] \end{aligned} \qquad \text{(A-16)}$$

$$\begin{aligned} y_3^{n+1} &= y_3^* + \left(1 - \tfrac{1}{\tau_\upsilon}\right) \sum_\alpha e_{\alpha y} f_\alpha^{neq}(\mathbf{r} - \mathbf{e}_\alpha \delta_t, (n+1)\Delta t) + F_{Ey} \delta_t \\ &= y_3^* - (\tau_\upsilon - 1) \sum_\alpha e_{\alpha y} \left[f_\alpha^{eq}(\mathbf{r} - \mathbf{e}_\alpha \delta_t, (n+1)\Delta t) - f_\alpha^{eq}(\mathbf{r} - 2\mathbf{e}_\alpha \delta_t, n\Delta t) \right] \\ &\quad - g\beta\delta_t (y_4^n - y_1^n T_m) \end{aligned} \qquad \text{(A-17)}$$

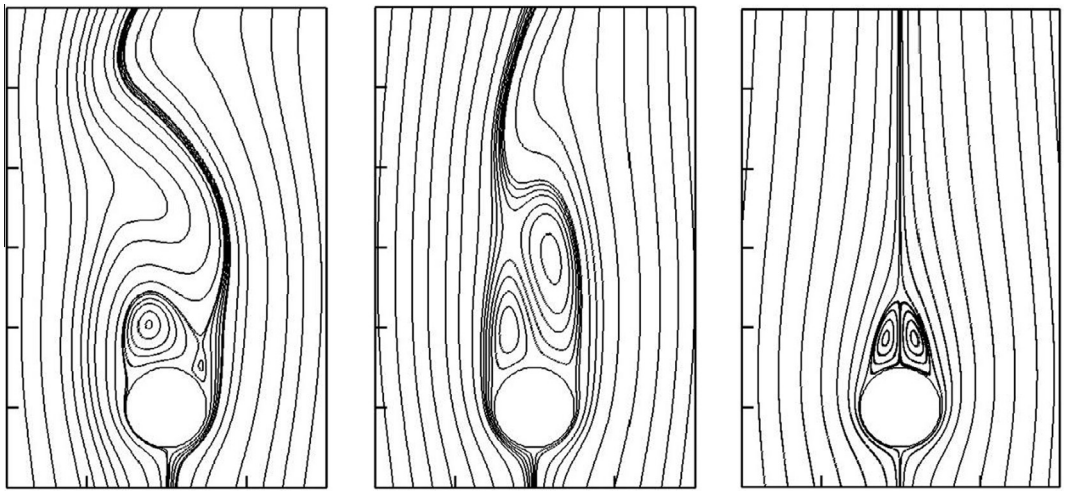


Fig. 11. Streamlines of mixed heat transfer at Re = 100 for different Richardson numbers. From left to right: Ri = 0, 0.1 and 0.5.

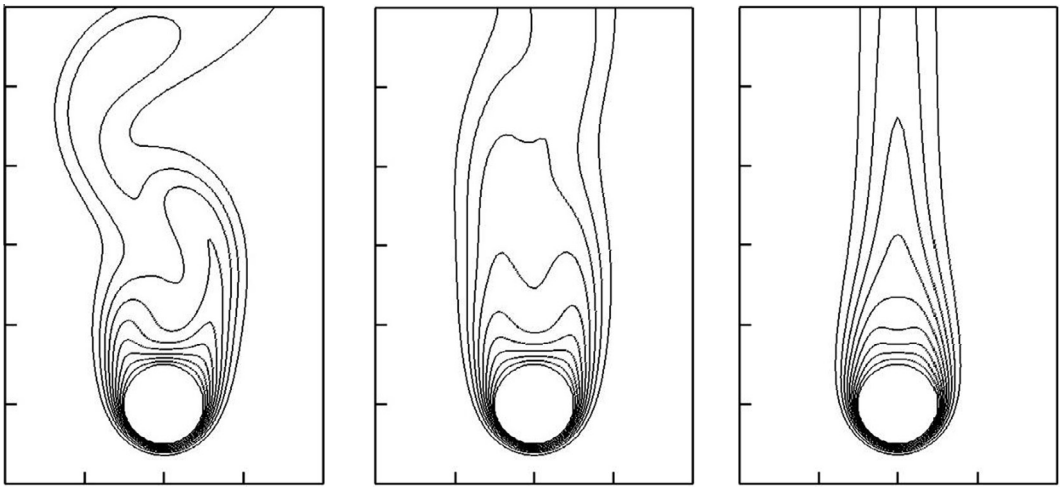


Fig. 12. Isotherms of mixed heat transfer at Re = 100 and different Richardson numbers. From left to right: Ri = 0, 0.1 and 0.5

Table 7
Comparison of the average Nusselt number and the Strouhal number of the mixed convection at $R_i = 0$ and Re = 100.

	$\overline{Nu}$	St
Hattom et al. [57], (experiment)	5.122	–
Williamson [58], (experiment)	–	0.163
Shi et al. [59]	5.142	0.164
Bhattacharyya and Singh [60]	5.114	0.166
Wang et al. [37]	5.137	0.163
Present	5.115	0.164

$$\begin{aligned} y_4^{n+1} &= y_4^* + \left(1 - \tfrac{1}{\tau_c}\right)\sum_\alpha g_\alpha^{neq}(\mathbf{r} - \mathbf{e}_\alpha\delta_t, (n+1)\Delta t) \\ &= y_4^* - (\tau_c - 1)\sum_\alpha \left[g_\alpha^{eq}(\mathbf{r} - \mathbf{e}_\alpha\delta_t, (n+1)\Delta t) - g_\alpha^{eq}(\mathbf{r} - 2\mathbf{e}_\alpha\delta_t, n\Delta t)\right] \\ &= y_4^* - (\tau_c - 1)(G - H) \end{aligned} \tag{A-18}$$

where G and H are in the following forms

$$G = \sum_\alpha g_\alpha^{eq}(\mathbf{r} - \mathbf{e}_\alpha\delta_t, (n+1)\Delta t) \tag{A-19}$$

$$H = \sum_\alpha g_\alpha^{eq}(\mathbf{r} - 2\mathbf{e}_\alpha\delta_t, n\Delta t) \tag{A-20}$$

For illustration purposes, we perform stability analysis of Eq. (A-18). In order to linearize the formulations, the terms G and H are differentiated as

$$\overline{\mathrm{d}G} = \left[\frac{\partial G}{\partial y_1}\right]^* \overline{\mathrm{d}y_1^*} + \left[\frac{\partial G}{\partial y_2}\right]^* \overline{\mathrm{d}y_2^*} + \left[\frac{\partial G}{\partial y_3}\right]^* \overline{\mathrm{d}y_3^*} + \left[\frac{\partial G}{\partial y_4}\right]^* \overline{\mathrm{d}y_4^*} \tag{A-21}$$

$$\overline{\mathrm{d}H} = \left[\frac{\partial H}{\partial y_1}\right]^n \overline{\mathrm{d}y_1^n} + \left[\frac{\partial H}{\partial y_2}\right]^n \overline{\mathrm{d}y_2^n} + \left[\frac{\partial H}{\partial y_3}\right]^n \overline{\mathrm{d}y_3^n} + \left[\frac{\partial H}{\partial y_4}\right]^n \overline{\mathrm{d}y_4^n} \tag{A-22}$$

Through a similar process as shown in Eqs. (A-9)–(A-12), we can derive

$$\left[\frac{\partial G}{\partial y_1}\right]^* = 0 \quad \left[\frac{\partial G}{\partial y_2}\right]^* = 0 \quad \left[\frac{\partial G}{\partial y_3}\right]^* = 0 \quad \left[\frac{\partial G}{\partial y_4}\right]^* = 1 \tag{A-23}$$

$$\left[\frac{\partial H}{\partial y_1}\right]^n = 0 \quad \left[\frac{\partial H}{\partial y_2}\right]^n = 0 \quad \left[\frac{\partial H}{\partial y_3}\right]^n = 0 \quad \left[\frac{\partial H}{\partial y_4}\right]^n = 1 \tag{A-24}$$

So, we have

$$\overline{\mathrm{d}G} = \overline{\mathrm{d}y_4^*} \quad \overline{\mathrm{d}H} = \overline{\mathrm{d}y_4^n} \tag{A-25}$$

Accordingly, differentiating Eq. (A-18) and substituting the relationships given by Eqs. (A-14) and (A-25), yields

$$\overline{\mathrm{d}y}_4^{n+1} = \overline{\mathrm{d}y}_4^{*} - (\tau_c - 1)\left(\overline{\mathrm{d}y}_4^{*} - \overline{\mathrm{d}y}_4^{n}\right) = \overline{\mathrm{d}y}_4^{n} \tag{A-26}$$

Following the process shown in Eqs. (A-9)–(A-26), we can also derive

$$\begin{aligned}\overline{\mathrm{d}y}_1^{n+1} &= \overline{\mathrm{d}y}_1^{n} \quad \overline{\mathrm{d}y}_2^{n+1} = \overline{\mathrm{d}y}_2^{n} \quad \overline{\mathrm{d}y}_3^{n+1} \\ &= g\beta T_m \delta_t \overline{\mathrm{d}y}_1^{n} + \overline{\mathrm{d}y}_3^{n} - g\beta\delta_t \overline{\mathrm{d}y}_4^{n}\end{aligned} \tag{A-27}$$

The above relationships can be written in a matrix form as shown in Eq. (78). By analyzing the eigenvalues of the characteristic matrix, we can prove that the simplified thermal lattice Boltzmann method is unconditionally stable.

References

[1] S. Chen, G.D. Doolen, Lattice Boltzmann method for fluid flows, Ann. Rev. Fluid Mech. 30 (1998) 329–364.
[2] C.K. Aidun, J.R. Clausen, Lattice-Boltzmann method for complex flows, Ann. Rev. Fluid Mech. 42 (2010) 439–472.
[3] S. Chen, H. Chen, D. Martnez, W. Matthaeus, Lattice Boltzmann model for simulation of magnetohydrodynamics, Phys. Rev. Lett. 67 (1991) 3776.
[4] Y. Qian, D. d'Humières, P. Lallemand, Lattice BGK models for Navier–Stokes equation, EPL (Europhys. Lett.) 17 (1992) 479.
[5] G. Tang, W. Tao, Y. He, Lattice Boltzmann method for gaseous microflows using kinetic theory boundary conditions, Phys. Fluids (1994-present) 17 (2005) 058101.
[6] X. He, S. Chen, R. Zhang, A lattice Boltzmann scheme for incompressible multiphase flow and its application in simulation of Rayleigh–Taylor instability, J. Comput. Phys. 152 (1999) 642–663.
[7] X. Shan, H. Chen, Lattice Boltzmann model for simulating flows with multiple phases and components, Phys. Rev. E 47 (1993) 1815.
[8] T. Inamuro, T. Ogata, S. Tajima, N. Konishi, A lattice Boltzmann method for incompressible two-phase flows with large density differences, J. Comput. Phys. 198 (2004) 628–644.
[9] M.R. Swift, W. Osborn, J. Yeomans, Lattice Boltzmann simulation of nonideal fluids, Phys. Rev. Lett. 75 (1995) 830.
[10] A.K. Gunstensen, D.H. Rothman, S. Zaleski, G. Zanetti, Lattice Boltzmann model of immiscible fluids, Phys. Rev. A 43 (1991) 4320.
[11] Y. Hu, D. Li, S. Shu, X. Niu, Immersed boundary-lattice Boltzmann simulation of natural convection in a square enclosure with a cylinder covered by porous layer, Int. J. Heat Mass Transfer 92 (2016) 1166–1170.
[12] U. Projahn, H. Rieger, H. Beer, Numerical analysis of laminar natural convection between concentric and eccentric cylinders, Numer. Heat Transfer 4 (1981) 131–146.
[13] T. Kuehn, R. Goldstein, An experimental and theoretical study of natural convection in the annulus between horizontal concentric cylinders, J. Fluid Mech. 74 (1976) 695–719.
[14] H.F. Oztop, E. Abu-Nada, Numerical study of natural convection in partially heated rectangular enclosures filled with nanofluids, Int. J. Heat Fluid Flow 29 (2008) 1326–1336.
[15] J. Douglas, Jim, T.F. Russell, Numerical methods for convection-dominated diffusion problems based on combining the method of characteristics with finite element or finite difference procedures, SIAM J. Numer. Anal. 19 (1982) 871–885.
[16] M. Hortmann, M. Perić, G. Scheuerer, Finite volume multigrid prediction of laminar natural convection: Bench - mark solutions, Int. J. Numer. Methods Fluids 11 (1990) 189–207.
[17] P. Le Quéré, Accurate solutions to the square thermally driven cavity at high Rayleigh number, Comput. Fluids 20 (1991) 29–41.
[18] J.T. Holdeman, J.W. Kim, Computation of incompressible thermal flows using Hermite finite elements, Comput. Methods Appl. Mech. Eng. 199 (2010) 3297–3304.
[19] M. Wang, J. He, J. Yu, N. Pan, Lattice Boltzmann modeling of the effective thermal conductivity for fibrous materials, Int. J. Therm. Sci. 46 (2007) 848–855.
[20] J. Wang, M. Wang, Z. Li, A lattice Boltzmann algorithm for fluid–solid conjugate heat transfer, Int. J. Therm. Sci. 46 (2007) 228–234.
[21] A. D'Orazio, M. Corcione, G.P. Celata, Application to natural convection enclosed flows of a lattice Boltzmann BGK model coupled with a general purpose thermal boundary condition, Int. J. Therm. Sci. 43 (2004) 575–586.
[22] R. Zhang, H. Chen, Lattice Boltzmann method for simulations of liquid-vapor thermal flows, Phys. Rev. E 67 (2003) 066711.
[23] W. Zhang, Z. Chai, Z. Guo, B. Shi, Lattice Boltzmann study of flow and temperature structures of non-isothermal laminar impinging streams, Commun. Comput. Phys. 13 (2013) 835–850.
[24] G. McNamara, B. Alder, Analysis of the lattice Boltzmann treatment of hydrodynamics, Physica A: Statist. Mech. Appl. 194 (1993) 218–228.
[25] F.J. Alexander, S. Chen, J. Sterling, Lattice boltzmann thermohydrodynamics, Phys. Rev. E 47 (1993) R2249.
[26] X. Shan, Simulation of Rayleigh-Bénard convection using a lattice Boltzmann method, Phys. Rev. E 55 (1997) 2780.
[27] A. Bartoloni, C. Battista, S. Cabasino, P. Paolucci, J. Pech, R. Sarno, G. Todesco, M. Torelli, W. Tross, P. Vicini, LBE simulations of Rayleigh–Benard convection on the APE100 parallel processor, Int. J. Mod. Phys. C 4 (1993) 993–1006.
[28] X. He, S. Chen, G.D. Doolen, A novel thermal model for the lattice Boltzmann method in incompressible limit, J. Comput. Phys. 146 (1998) 282–300.
[29] J. Onishi, Y. Chen, H. Ohashi, Lattice Boltzmann simulation of natural convection in a square cavity, JSME Int. J. Ser. B Fluids Therm. Eng. 44 (2001) 53–62.
[30] A. d'Orazio, S. Succi, C. Arrighetti, Lattice Boltzmann simulation of open flows with heat transfer, Phys. Fluids (1994-present) 15 (2003) 2778–2781.
[31] A. Karimipour, A.H. Nezhad, A. D'Orazio, E. Shirani, The effects of inclination angle and Prandtl number on the mixed convection in the inclined lid driven cavity using lattice Boltzmann method, J. Theor. Appl. Mech. 51 (2013) 447–462.
[32] A. Karimipour, A.H. Nezhad, A. D'Orazio, E. Shirani, Investigation of the gravity effects on the mixed convection heat transfer in a microchannel using lattice Boltzmann method, Int. J. Therm. Sci. 54 (2012) 142–152.
[33] Y. Peng, C. Shu, Y. Chew, Simplified thermal lattice Boltzmann model for incompressible thermal flows, Phys. Rev. E 68 (2003) 026701.
[34] A. D'Orazio, A. Karimipour, A.H. Nezhad, E. Shirani, Lattice Boltzmann method with heat flux boundary condition applied to mixed convection in inclined lid driven cavity, Meccanica 50 (2015) 945–962.
[35] A.E.F. Monfared, A. Sarrafi, S. Jafari, M. Schaffie, Thermal flux simulations by lattice Boltzmann method; investigation of high Richardson number cross flows over tandem square cylinders, Int. J. Heat Mass Transfer 86 (2015) 563–580.
[36] J.D. Sterling, S. Chen, Stability analysis of lattice Boltzmann methods, J. Comput. Phys. 123 (1996) 196–206.
[37] Y. Wang, C. Shu, C. Teo, Thermal lattice Boltzmann flux solver and its application for simulation of incompressible thermal flows, Comput. Fluids 94 (2014) 98–111.
[38] Y. Wang, L. Yang, C. Shu, From lattice Boltzmann method to lattice Boltzmann flux solver, Entropy 17 (2015) 7713–7735.
[39] F.M. White, Fluid Mechanics, McGraw-Hill, New York, 2003.
[40] Y.A. Cengel, A.J. Ghajar, H. Ma, Heat and Mass Transfer: Fundamentals & Applications, 4 ed., McGraw-Hill, 2011.
[41] Z. Guo, C. Shu, Lattice Boltzmann Method and its Applications in Engineering, World Scientific, 2013.
[42] S. Succi, The Lattice Boltzmann Equation: For Fluid Dynamics and Beyond, Oxford university press, 2001.
[43] R. Benzi, S. Succi, M. Vergassola, The lattice Boltzmann equation: theory and applications, Phys. Rep. 222 (1992) 145–197.
[44] U. Frisch, D. d'Humieres, B. Hasslacher, P. Lallemand, Y. Pomeau, J.-P. Rivet, Lattice gas hydrodynamics in two and three dimensions, Complex Syst. 1 (1987) 649–707.
[45] T. Inamuro, M. Yoshino, F. Ogino, Accuracy of the lattice Boltzmann method for small Knudsen number with finite Reynolds number, Phys. Fluids (1994-present) 9 (1997) 3535–3542.
[46] J. Kim, P. Moin, Application of a fractional-step method to incompressible Navier–Stokes equations, J. Comput. Phys. 59 (1985) 308–323.
[47] C. Shu, Y. Wang, C. Teo, J. Wu, Development of lattice Boltzmann flux solver for simulation of incompressible flows, Adv. Appl. Math. Mech. 6 (2014) 436–460.
[48] J.D. Anderson, J. Wendt, Computational Fluid Dynamics, Springer, 1995.
[49] Z. Guo, C. Zheng, B. Shi, An extrapolation method for boundary conditions in lattice Boltzmann method, Phys. Fluids (1994-present) 14 (2002) 2007–2010.
[50] C. Shu, H. Xue, Comparison of two approaches for implementing stream function boundary conditions in DQ simulation of natural convection in a square cavity, Int. J. Heat Fluid Flow 19 (1998) 59–68.
[51] G. de Vahl Davis, Natural convection of air in a square cavity: a bench mark numerical solution, Int. J. Numer. Methods Fluids 3 (1983) 249–264.
[52] D.A. Mayne, A.S. Usmani, M. Crapper, h-adaptive finite element solution of high Rayleigh number thermally driven cavity problem, Int. J. Numer. Methods Heat Fluid Flow 10 (2000) 598–615.
[53] D. Contrino, P. Lallemand, P. Asinari, L.-S. Luo, Lattice-Boltzmann simulations of the thermally driven 2D square cavity at high Rayleigh numbers, J. Comput. Phys. 275 (2014) 257–272.
[54] C. Shu, Application of differential quadrature method to simulate natural convection in a concentric annulus, Int. J. Numer. Methods Fluids 30 (1999) 977–993.
[55] H. Badr, Laminar combined convection from a horizontal cylinder-parallel and contra flow regimes, Int. J. Heat Mass Transfer 27 (1984) 15–27.
[56] K.-S. Chang, J.-Y. Sa, Numerical study of the unsteady mixed convection heat transfer from a circular cylinder, Int. Commun. Heat Mass Transfer 16 (1989) 427–434.
[57] A. Hatton, D. James, H. Swire, Combined forced and natural convection with low-speed air flow over horizontal cylinders, J. Fluid Mech. 42 (1970) 17–31.
[58] C. Williamson, Oblique and parallel modes of vortex shedding in the wake of a circular cylinder at low Reynolds numbers, J. Fluid Mech. 206 (1989) 579–627.
[59] J.-M. Shi, D. Gerlach, M. Breuer, G. Biswas, F. Durst, Heating effect on steady and unsteady horizontal laminar flow of air past a circular cylinder, Phys. Fluids (1994-present) 16 (2004) 4331–4345.
[60] S. Bhattacharyya, A. Singh, Vortex shedding and heat transfer dependence on effective Reynolds number for mixed convection around a cylinder in cross flow, Int. J. Heat Mass Transfer 53 (2010) 3202–3212.